The Recovery–Certification Gap: A Theory of Adversarial Unlearning

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Abstract

This paper studies machine-unlearning robustness through the per-example *recovery radius*: the smallest input perturbation that makes an unlearned model predict a forgotten class again. Recovery-radius distributions are oracle-free to measure and expose failures that clean forget accuracy can miss; when a retain-only oracle is available, it calibrates whether a forget-vs-retain gap reflects residual forgotten knowledge rather than generic target-label fragility. Four results organize the study on a controlled benchmark (primarily CIFAR-10 ViT checkpoints, with single-seed CIFAR-100 cross-checks). First, a recovery–certification gap gives an audit-estimable necessary condition for certificates that bound predictive behavior—an axis that mainstream parameter-distribution certified-unlearning guarantees, which certify indistinguishability of the weight distribution from retraining, provably leave open. Second, a Pinsker-style selective-leakage bound interprets large forget-vs-retain recovery-rate gaps while a larger-n retain-only oracle audit separates residual knowledge from baseline asymmetry; a benchmark-wide audit (132 rows, 57 checkpoints) then uncovers a fragility confound in this gap—globally brittle checkpoints inflate apparent selectivity through a super-linear forget-radius elasticity (b=1.31, confirmed causally as $\hat{b}_{\text{causal}}$=1.36, CI $[1.31, 1.84]$, by varying base input-space smoothness with the unlearning method held fixed)—and a fragility-matched taxonomy isolates the methods that genuinely leak (SalUn, RURK, smoothed-margin) from those whose "leak" is brittleness (SCRUB, at or below the oracle null). Third, a covering-family impossibility shows that exact worst-case invariance over all local delivery surfaces would force a nontrivial balanced classifier to be constant, so practical defenses must target distributional rather than global worst-case transfer. Fourth, a Gaussian-smoothed recovery objective yields a post-training per-input input-space lower bound for a binary forgotten-class detector; the corresponding native L_2 deterministic recovery audit reports radii of 0.46–0.57 across three seeds against a detector-radius scale of $R = 0.193$, while a smoothing-scale sweep exposes a certification–utility tradeoff. Code, result JSONs, and selected checkpoints are available at `https://github.com/DevT02/ForgetGate`.

1 Introduction

Machine unlearning aims to remove the influence of specified training data from a trained model, but clean forget accuracy is an incomplete audit. A model can fail to predict the forgotten class on unperturbed inputs while still recovering that class under a small targeted perturbation. The central object in this paper is therefore the *recovery radius*: the minimum perturbation required to

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re-elicit the forgotten class. This radius distribution, together with recovery rates at fixed budgets, is the quantity that an operational audit and any input-space certificate should address.

The study evaluates eight unlearning rows under a primary recovery-radius battery—per-example conditional recovery, conditional patches, border-frame delivery, and multi-patch delivery—with universal perturbations, amortized short-budget attacks, prompt/probe controls, and AWP used as auxiliary probes. No attack or defense is universal. Some methods are vulnerable to targeted per-example recovery but not patches; others look robust until the delivery surface changes; and defenses that close one channel often leave another open.

This framing separates attack success from selective leakage. When a perturbation recovers the forgotten class, it may expose residual class-specific information, or it may merely induce a generic adversarial target-label failure. The audit therefore compares forget-set examples with matched retain-class controls, and, when available, with a retain-only oracle null: a model retrained without the forgotten class. Recovery radius remains the oracle-free primitive; the retain-only oracle calibrates whether a forget-vs-retain gap is residual knowledge rather than baseline target-class asymmetry.

The positive defense results are modest. A post-hoc feature-subspace defense gives positive point-estimate shifts on BalDRO, SalUn, and ORBIT without hurting clean accuracy in the measured rows, with the clearest practical signal on ORBIT and SalUn and only a tiny BalDRO shift; RURK and SCRUB are largely unaffected. Patch-aware training partially closes the patch channel on ORBIT and CE-U, but does not transfer reliably to frame or multi-patch delivery. The benchmark code, result JSONs, and selected checkpoints are available at `https://github.com/DevT02/ForgetGate`.

The primary and auxiliary attack regimes measure the same object, and that object ties red-team auditing to certified unlearning. Section 3 gives four organizing results. Theorem 1 bounds recovery radius below by the target margin and a Lipschitz constant, giving an audit-estimable necessary condition for certified-unlearning budgets rather than a strict falsification test. Theorem 2 gives a variational-information interpretation of large forget-vs-retain recovery-rate gaps; the retain-only oracle null is the empirical calibration used later to identify residual knowledge. Theorem 3 shows that exact worst-case robustness over a coordinate-covering delivery family forces constancy, which motivates distributional local-delivery defenses instead of global worst-case invariance. Theorem 4 attaches a per-input input-space randomized-smoothing lower bound to a binary forgotten-class detector. Section 8 checks these claims through the Lipschitz–radius diagnostic, the Pinsker/oracle-null audit, the patch/frame/multi-patch transfer results, and the smoothed-margin recovery-radius audits.

2 Threat Models and Attack Regimes

2.1 Universal perturbations

The first regime includes Fourier, tile, warp, and codebook-style universal perturbations. These attacks are useful primarily as a negative result: on stronger methods, the residual vulnerability is not well-compressed into one shared low-complexity direction.

2.2 Per-example conditional recovery

The main broad attack optimizes the forgotten-class target margin

$$m_t(x) = z_t(x) - \max_{j \neq t} z_j(x), \tag{1}$$

using strong first-order optimizers such as `adam_margin` and `apgd_margin`. This regime asks whether a forgotten behavior is locally recoverable by a targeted per-example perturbation inside a fixed norm budget.

2.3 Amortized short-budget attacks

A learned image-conditional perturbation generator is trained from short Adam teacher trajectories and refined with a tiny local budget. This attack family is not universal, but it improves matched low-step attacks on ORBIT and SalUn.

2.4 Conditional patches

The patch regime uses conditional small-area patches in which both patch content and placement depend on the input. This is a patch-based regime, not a universal trigger claim.

2.5 Additional local delivery surfaces

To test whether patch-aware defenses generalize beyond one square patch, the benchmark also includes conditional border-frame attacks and conditional multi-patch attacks. These are stronger local-delivery checks than a single-patch evaluation because they alter geometry while preserving the same selective red-team objective.

3 A Theory of Adversarial Unlearning

The empirical regimes of this paper measure a single object: the smallest input-space perturbation that re-elicits the forgotten class on a per-example basis. This section names that object, ties it quantitatively to certified unlearning, separates selective leakage from generic adversarial fragility, and gives a constructive impossibility result for the open "local-delivery transfer" problem. The benchmark sections that follow are used as empirical checks and diagnostics for these statements rather than as standalone attack curves.

3.1 Setup and the selective recovery gap

Notation. Let $\mathcal{X} = [0,1]^d$ be the input space and $\mathcal{Y} = [K]$ the label set. Fix the forget class $t \in [K]$. Let $D = D_R \cup D_F$ with $D_F = \{(x_i, t)\}_{i=1}^{n_f}$. Let $\mathcal{A}$ be an unlearning algorithm producing $\theta_u = \mathcal{A}(D, t)$. Let θ_o denote a retain-only oracle trained on D_R. Let $f_\theta : \mathcal{X} \to \mathbb{R}^K$ be the logit map and define the forget-class *target margin*

$$m_t(x;\theta) \;:=\; f_\theta(x)_t - \max_{j\neq t} f_\theta(x)_j. \tag{2}$$

The unlearned model predicts the forgotten class at x iff $m_t(x;\theta_u) > 0$.

Recovery radius. For an L_∞ budget,

$$r_t(x;\theta) \;:=\; \inf\{\epsilon \geq 0 \mid \exists\, x' \in \mathcal{X}, \|x' - x\|_\infty \leq \epsilon,\; m_t(x';\theta) > 0\}, \tag{3}$$

with $r_t(x;\theta) = 0$ when the clean prediction already matches. A per-example finite-budget audit returns an estimator $\hat{r}_t(x;\theta_u)$ via probe-then-binary search with PGD / APGD inner loops. Clean forgotten-class predictions are recorded as radius 0; successful attacks inside $\epsilon_{\max}$ receive a finite radius estimate; failures are right-censored as $r_t(x;\theta_u) > \epsilon_{\max}$. Consequently, empirical tables distinguish recovery rate at $\epsilon_{\max}$, censored all-sample medians, and success-conditioned medians. The theoretical statements below concern the all-sample radius distribution; success-conditioned medians are auxiliary descriptive statistics.

The right population object. Let μ_t denote the forget-class input distribution and $\mu_{\neg t}$ the matched non-forget control distribution used throughout the paper. Define the *selective recovery gap* at scale ϵ:

$$\boxed{\mathcal{R}_\epsilon(\theta_u) \;:=\; \Pr_{x\sim\mu_t}\big[r_t(x;\theta_u) \le \epsilon\big] \;-\; \Pr_{x\sim\mu_{\neg t}}\big[r_t(x;\theta_u) \le \epsilon\big].} \tag{4}$$

$\mathcal{R}_\epsilon \in [-1,1]$ is the signed difference between forget-side and retain-side recoverability at audit scale ϵ. The forget–retain gap reported throughout the paper is exactly the empirical estimator of $\mathcal{R}_\epsilon$. Two qualitative regimes the paper invokes informally become precise statements about $\mathcal{R}_\epsilon$:

- **Selective leakage** (large forget-vs-retain recovery gap): $\mathcal{R}_\epsilon(\theta_u) > 0$ — forget inputs are strictly easier to recover than retain controls.
- **Generic fragility** (near-zero forget-vs-retain gap): $\mathcal{R}_\epsilon(\theta_u) \approx 0$ — the model is adversarially fragile, but not selectively so on the forget class.

3.2 Theorem 1: the recovery–certification gap

This bridges the per-example recovery radius to any *certified-unlearning* budget the method might claim.

Theorem 1 (Recovery–certification gap)**.** *Suppose the target-margin function is L-Lipschitz in $\|\cdot\|_\infty$ over $\mathcal{X}$, i.e. $|m_t(x;\theta_u) - m_t(x';\theta_u)| \le L\,\|x-x'\|_\infty$ for all $x, x' \in \mathcal{X}$. Then for every $x \in \mathcal{X}$,*

$$r_t(x;\theta_u) \;\ge\; \frac{-m_t(x;\theta_u)}{L}. \tag{5}$$

Equivalently,

$$\Pr_{x\sim\mu_t}\big[r_t(x;\theta_u) \le \epsilon\big] \;\le\; \Pr_{x\sim\mu_t}\big[\,m_t(x;\theta_u) \ge -L\epsilon\,\big]. \tag{6}$$

Proof. For any $\epsilon \ge 0$ and any $x' \in B_\infty(x,\epsilon) \cap \mathcal{X}$, the Lipschitz bound gives $m_t(x';\theta_u) \le m_t(x;\theta_u) + L\epsilon$. For the recovery event $m_t(x';\theta_u) > 0$ to hold we therefore require $\epsilon > -m_t(x;\theta_u)/L$, which proves (5). Then (6) follows by integrating against μ_t. □

Theorem 1 upper-bounds the recovery probability through the margin; the audit supplies the matching *lower* bound, and—unlike the Lipschitz proxy—it requires no assumption on the inner optimizer.

Corollary 1 (Attack-observable recovery lower bound)**.** *Suppose a finite-budget audit exhibits, for an α fraction of sampled forget inputs, an explicit perturbation x' with $\|x'-x\|_\infty \le \rho$ and $m_t(x';\theta_u) > 0$. Then, with no assumption that the optimizer is complete or that $\hat{r}_t$ is exact,*

$$\Pr_{x\sim\mu_t}\big[r_t(x;\theta_u) \le \rho\big] \;\ge\; \alpha \quad \textit{(in expectation over the sampled forget inputs).} \tag{7}$$

Proof. Each witnessed perturbation satisfies the recovery event at distance $\le \rho$, so by the infimum definition $r_t(x;\theta_u) \le \rho$ for that input—a one-sided fact that an incomplete optimizer cannot violate. The empirical recovery frequency at scale ρ is therefore an unbiased estimator of a quantity that is itself $\le \Pr_{\mu_t}[r_t \le \rho]$, since every audit failure is genuinely right-censored above ρ rather than miscounted. □

This is the clean form of the audit's evidentiary status: successful attacks are valid upper bounds on the true radius and failures are honest right-censorings, so the measured recovery rate is attack-valid without claiming attack optimality. Theorem 1 and Corollary 1 sandwich the population recovery probability $\Pr_{\mu_t}[r_t \le \epsilon]$ between the observed audit rate (below) and the Lipschitz–margin bound (above).

Corollary 2 (Certified unlearning needs separated margins)**.** *Instantiate Theorem 1 with the* probability margin $\bar{m}_t(x;\theta) := \pi_\theta(x)_t - \max_{j\neq t}\pi_\theta(x)_j \in [-1,1]$, *where* $\pi_\theta = \mathrm{softmax}\circ f_\theta$. *Because softmax is strictly monotone,* $\bar{m}_t$ *has the same sign as the logit margin and defines the same recovery event; let* L *be its input-space Lipschitz constant. Let* θ_u *be* (ϵ_c,δ)*-certified relative to the oracle* θ_o *in the conditional TV sense,*

$$\Pr_{x\sim\mu_t}\Big[\mathrm{TV}\big(\pi_{\theta_u}(\cdot\mid x),\,\pi_{\theta_o}(\cdot\mid x)\big)\le\epsilon_c\Big]\ \ge\ 1-\delta,$$

and suppose the oracle is separated on average, $\mathbb{E}_{\mu_t}[-\bar{m}_t(x;\theta_o)]\ge\gamma>0$. *Then, whenever* $\gamma-2(\epsilon_c+\delta)>0$,

$$\mathbb{E}_{x\sim\mu_t}[\,r_t(x;\theta_u)\,]\ \ge\ \frac{\gamma-2(\epsilon_c+\delta)}{L}. \tag{8}$$

If the numerator is non-positive, the lower bound is vacuous.

Proof. For distributions p,q on $[K]$ and any class k, the event $A=\{k\}$ gives $|p_k-q_k| = |p(A)-q(A)| \le \mathrm{TV}(p,q)$, and $a\mapsto\max_{j\neq t}a_j$ is 1-Lipschitz in the sup-norm. Hence pointwise,

$$\begin{aligned}|\bar{m}_t(x;\theta_u)-\bar{m}_t(x;\theta_o)| &\le |\pi_{\theta_u}(x)_t-\pi_{\theta_o}(x)_t| + |\max_{j\neq t}\pi_{\theta_u}(x)_j-\max_{j\neq t}\pi_{\theta_o}(x)_j| \\ &\le 2\,\mathrm{TV}\big(\pi_{\theta_u}(\cdot\mid x),\pi_{\theta_o}(\cdot\mid x)\big).\end{aligned}$$

Since the TV distance is bounded by ϵ_c with probability $1-\delta$ and by 1 otherwise, its expectation is at most $\epsilon_c(1-\delta)+\delta\le\epsilon_c+\delta$. Taking expectations under μ_t gives $\mathbb{E}[-\bar{m}_t(\cdot;\theta_u)]\ge\gamma-2(\epsilon_c+\delta)$; applying Theorem 1 to $\bar{m}_t$ (pointwise $r_t(x)\ge-\bar{m}_t(x)/L$) yields the claim. The bounded margin $\bar{m}_t\in[-1,1]$ removes any dependence on a logit bound. □

What this says about the benchmark. Every method in the paper has a measured $\hat{r}$ distribution and a straightforwardly estimable Lipschitz proxy $\hat{L}$ (e.g. the spectral norm of the Jacobian of the margin at audit points, which an existing Jacobian-access audit already computes), used here as a logit-margin proxy for the probability-margin constant L of Corollary 2. Corollary 2 contrapositively gives a *theoretical floor on the certifiable unlearning budget*:

$$\epsilon_c+\delta\ \ge\ \frac{\gamma-L\cdot\mathbb{E}[\hat{r}]}{2},$$

estimated per method from its recovery-radius audit; the scatter of Section 8 plots the more robust median $\hat{r}$ as a stand-in for the mean. Two gaps separate $\hat{L}$ from the constant Corollary 2 requires. First, a *norm* gap: the L_∞-Lipschitz constant of the scalar margin is controlled by the dual norm $\|\nabla m\|_1$, whereas the spectral Jacobian proxy reports $\|\nabla m\|_2\le\|\nabla m\|_1$ and can therefore understate it by up to a $\sqrt{d}$ factor; a tighter audit would estimate $\|\nabla m\|_1$ directly. Second, a *locality* gap: $\hat{L}$ is a local proxy for the (NP-hard) global Lipschitz constant, and local proxies on ViTs tend to under-estimate it. Both gaps push the heuristic floor estimate the same way—they inflate it and could in principle produce a false rejection—so this is treated as an audit-estimable *necessary condition* rather than a strict falsification test: a method whose median recovery radius is small relative to the oracle margin is unlikely to be honestly marketable at small ϵ_c, and the estimate serves as a heuristic ranking and baseline reality-check that audits with no access to the oracle distribution can already perform.

Two certificate types, and which one the radius probes. Corollary 2 consumes a *conditional-predictive* budget: a bound on the per-input TV distance between the deployed model's class distribution $\pi_{\theta_u}(\cdot \mid x)$ and the oracle's. This is deliberately *not* the object that the mainstream certified-unlearning literature provides. Langevin unlearning [10], Rewind-to-Delete [20], the neural-network certificates of [21], and retain-sensitivity calibration [19] all certify a *parameter-distribution* (ϵ, δ): statistical indistinguishability, in the differential-privacy sense, between the *distribution* over unlearned weights and the retrain-from-scratch distribution. A recent verification survey [22] draws the same line—parameter-distribution certificates leave individual prediction stability and input-perturbation resurfacing at the forgotten point uncertified. Converting a parameter-distribution (ϵ, δ) into the conditional-predictive TV that Corollary 2 requires is not free: it needs the model to be Lipschitz in its parameters and certifies the *distribution* over realized models, not the single deployed checkpoint an attacker actually faces. The recovery radius therefore measures exactly the axis these certificates do not bound—whether one deployed model resurfaces the forgotten class under a small input perturbation—so Corollary 2 is read not as a falsification of existing certificates (which make a different claim) but as the floor a *predictive-type* certificate would have to respect, and as the formal reason the recovery–certification gap is structural rather than incidental. Where a method does expose a usable predictive (ϵ_c, δ), the same inequality converts it into a recovery-radius lower bound [10, 6, 24, 19].

3.3 Theorem 2: selective leakage and oracle-null calibration

Theorem 2 (Pinsker bound on selective leakage)**.** *Let* $p_F := \Pr_{\mu_t}[r_t \le \epsilon]$ *and* $p_R := \Pr_{\mu_{\neg t}}[r_t \le \epsilon]$*. Then*

$$|\mathcal{R}_\epsilon(\theta_u)| \;=\; |p_F - p_R| \;\le\; \sqrt{\tfrac{1}{2}\,\mathrm{KL}\big(\mathrm{Ber}(p_F)\,\|\,\mathrm{Ber}(p_R)\big)}. \tag{9}$$

Consequently, $|\mathcal{R}_\epsilon(\theta_u)| \ge c$ *forces* $\mathrm{KL}\big(\mathrm{Ber}(p_F)\|\mathrm{Ber}(p_R)\big) \ge 2c^2$*, i.e. the audit's binary recoverability outcome distinguishes forget from retain at variational rate* c*.*

Proof. Pinsker's inequality applied to the two Bernoulli laws on the binary outcome $\mathbf{1}\{r_t \le \epsilon\}$. □

Corollary 3 (Operational read of the gap)**.** *With equal priors over forget and matched-control inputs, the Bayes-optimal classifier that observes only the binary recovery event* $\mathbf{1}\{r_t \le \epsilon\}$ *attains accuracy*

$$\tfrac{1}{2}\big(1 + |p_F - p_R|\big) \;=\; \tfrac{1}{2}\big(1 + |\mathcal{R}_\epsilon(\theta_u)|\big). \tag{10}$$

Proof. The total variation distance between $\mathrm{Ber}(p_F)$ and $\mathrm{Ber}(p_R)$ on the binary outcome is exactly $|p_F - p_R|$, and the Bayes accuracy of distinguishing two equiprobable hypotheses equals $\frac{1}{2}(1+\mathrm{TV})$. □

This gives the gap a direct operational meaning: a forget–retain recovery gap of $|\mathcal{R}_\epsilon|$ is exactly the advantage over chance with which an observer can decide "forget vs. retain" from the single recovery bit. The Pinsker statement above is then the secondary information-theoretic consequence.

Oracle-null calibration. The theorem concerns the raw forget–retain gap of one unlearned model. The empirical oracle null used in Section 8 is a separate calibration step. For a retain-only oracle θ_o, define

$$S_\epsilon(\theta_u, \theta_o) \;:=\; \big[p_F(\theta_u) - p_R(\theta_u)\big] \;-\; \big[p_F(\theta_o) - p_R(\theta_o)\big].$$

Theorem 2 gives a variational interpretation to the first bracket when it is large; the null subtraction checks whether the observed gap exceeds the baseline asymmetry of driving any input toward the forgotten class.

Corollary 4 (Near-zero gaps give no selective signal)**.** *If $\mathcal{R}_\epsilon(\theta_u) \to 0$ as $n_f \to \infty$ at some audit scale ϵ, no unlearning claim about μ_t-specific information removal is empirically supported by this recovery event at that scale: the data-conditional binary leakage event is indistinguishable from the retain-control baseline. This does not rule out other budgets, attacks, or leakage channels.*

Why this matters. The paper's qualitative observation that SCRUB usually behaves like generic adversarial fragility under this audit becomes a variational-information comparison (not a finite-sample hypothesis test; see the small-n caveat in the limitations): the binary recoverability outcomes on forget and retain controls are samples from $\mathrm{Ber}(p_F)$ and $\mathrm{Ber}(p_R)$, and Theorem 2 turns an observed gap into a *lower* bound on the KL it carries about the forget condition. The direction matters: a large measured gap *forces* a quantitative KL and is selective signal, whereas a near-zero gap leaves the KL unconstrained from above, so any generic-fragility reading rests on the direct equality of the measured rates and null comparison, not on Pinsker.

3.4 Forget-radius super-elasticity

Theorem 1 reads each radius through a *single* Lipschitz scale L. Taken literally, that single-scale reading makes a falsifiable prediction about how the forget and retain-control radii should co-move across checkpoints, and the prediction is wrong in a specific, informative direction. This subsection isolates the discrepancy and shows what it measures.

Fix a checkpoint with margin-Lipschitz constant L. Write μ_F for the forget population (target = the forgotten class) and μ_R for the retain-control population, and let $r_F = \mathrm{med}_{\mu_F} r_t$, $r_R = \mathrm{med}_{\mu_R} r_t$ be their median recovery radii. In the local-linear regime where Theorem 1 is tight, $r_t(x) \approx -m_t(x)/L$, so the medians satisfy $r_F \approx \tilde{m}_F/L$ and $r_R \approx \tilde{m}_R/L$, where $\tilde{m}_\bullet = \mathrm{med}_{\mu_\bullet}[-m_t(x)] > 0$ is the median *margin deficit* an attack must close.

Proposition 1 (Elasticity identity)**.** *Suppose that, across checkpoints, the retain-control deficit is fragility-insensitive, $\tilde{m}_R(L) = c_R$, while the forget deficit co-attenuates with the global scale as a power law, $\tilde{m}_F(L) = c_F\, L^{-\beta}$ for some $\beta \geq 0$. Then the log–log regression of the forget radius on the retain-control radius across checkpoints has slope*

$$b \;=\; \frac{\mathrm{d}\,\log r_F}{\mathrm{d}\,\log r_R} \;=\; 1+\beta \;\geq\; 1, \tag{11}$$

with equality iff $\beta = 0$.

Proof. From $r_R = c_R/L$ we get $\log L = \log c_R - \log r_R$. From $r_F = \tilde{m}_F(L)/L = c_F L^{-(1+\beta)}$ we get $\log r_F = \log c_F - (1+\beta)\log L$. Substituting, $\log r_F = [\log c_F - (1+\beta)\log c_R] + (1+\beta)\log r_R$, whose slope in $\log r_R$ is $1+\beta$. □

Interpretation. The single-scale reading of Theorem 1—margins fixed, only L varying across checkpoints—is exactly the case $\beta = 0$, which predicts $b = 1$: forget and retain radii would shrink in lockstep as a checkpoint becomes globally brittle, and the forget/retain *ratio* (hence apparent selectivity) would be scale-invariant. The benchmark instead exhibits $\hat{b} = 1.31$ (suite-clustered 95% CI $[1.11, 2.04]$, Section 8), identifying $\hat{\beta} = \hat{b} - 1 \approx 0.31 > 0$: the forget deficit is *not* fragility-invariant. As a checkpoint's representation becomes less smooth (larger L), the suppressed class loses margin deficit faster than the retain controls, so the forget radius collapses super-linearly in the global fragility scale. This is the mechanism behind the fragility confound of Section 8: because $b > 1$, part of any cross-checkpoint selectivity ranking is a fragility ranking, and only fragility-matched comparisons (residualizing on L, equivalently on r_R) isolate method-specific leakage.

What sets β. The exponent is a property of the forget *margin distribution*, not of the optimizer. When the suppressed-class deficits are heterogeneous—a spread of $-m_t$ straddling the recovery boundary rather than a uniform deep suppression—the median deficit sits on a steeper part of the deficit CDF, so a global compression of headroom moves the median more than one-for-one and yields $\beta > 0$. Uniform deep suppression (all forget logits driven far below threshold, e.g. the parity SCRUB seed whose $\log p_t < -37$ for every forget sample) flattens that CDF near the median and drives $\beta \to 0$, i.e. $b \to 1$ and no selective gap. This ties the elasticity to the measurable within-checkpoint deficit spread that Section 8 correlates with selectivity, and makes Proposition 1 a prediction rather than a restatement: checkpoints with more heterogeneous forget margins should show larger local b. We do not claim a closed form for β in terms of the deficit tail; the identity (11) only requires the power-law co-scaling, which the log–log linearity of the data (Section 8) supports empirically.

Scope. The slope $\hat{b} = 1.31$ above is *observational*: L varies with the method, so it pools method and fragility. A within-method controlled estimate—fixing the method and dialing a single robustness knob—identifies b causally. Section 3.9 reports such an estimate: varying base input-space smoothness (adversarial pre-training strength) while holding SalUn fixed yields $\hat{b}_{\text{causal}} = 1.36$ (95% CI $[1.31, 1.84]$), excluding $b = 1$ and matching the observational value (Figure 1D). Adapter-side weight decay, by contrast, is a first-stage null—the radius is governed by the base geometry, not the low-rank adapter—which is what identifies base smoothness as the correct instrument.

3.5 Theorem 3: local-delivery transfer impossibility

This formalizes the paper's open problem ("patch-aware defenses fail to transfer to frame or multi-patch delivery") as a structural impossibility.

Delivery families and coordinate coverage. A *delivery mask* is a subset $M \subseteq [d]$ of input coordinates. A *delivery family* $\mathcal{D} \subseteq 2^{[d]}$ is *coordinate-covering* if every input coordinate lies in at least one of its masks, $\bigcup_{M \in \mathcal{D}} M = [d]$. A delivery-conditional perturbation under $\mathcal{D}$ with budget ϵ is a $\delta \in \mathbb{R}^d$ with $\operatorname{supp}(\delta) \subseteq M$ for some $M \in \mathcal{D}$ and $\|\delta\|_\infty \le \epsilon$. This is a strong worst-case support-subset model: any subset of a delivery mask, including a single coordinate, is admissible. Structured visible patches or sampled mask distributions are weaker average-case settings and are not ruled out by the theorem.

Lemma 1 (Patches alone are coordinate-covering)**.** *For 224×224 images on $[0,1]^d$, the family of all 32×32 square delivery masks at every position is coordinate-covering; adjoining the 16-pixel border frames leaves it coordinate-covering.*

Proof. Every pixel lies inside at least one 32×32 window, so the union of the patch masks is all of $[d]$; adding more masks only enlarges the union. □

Theorem 3 (Budget-graded local-delivery transfer impossibility)**.** *Let $\mathcal{D}$ be a coordinate-covering delivery family and let $f_\theta : [0,1]^d \to \mathcal{Y}$ be any classifier. Fix* any *budget $\epsilon > 0$ and suppose f_θ is* worst-case invariant *to all delivery-conditional perturbations under $\mathcal{D}$ at budget ϵ:*

$$\forall x,\ \forall M \in \mathcal{D},\ \forall \delta \text{ with } \operatorname{supp}(\delta) \subseteq M,\ \|\delta\|_\infty \le \epsilon,\ x + \delta \in [0,1]^d : \quad f_\theta(x+\delta) = f_\theta(x). \tag{12}$$

Then f_θ is constant on $[0,1]^d$.

Proof. Fix $x, x' \in [0,1]^d$; we connect them by admissible ϵ-steps, one coordinate at a time. For coordinate i, choose a mask $M_i \in \mathcal{D}$ with $i \in M_i$ (coordinate coverage). Move the i-th entry monotonically from its current value to x'_i in increments of magnitude at most ϵ; each increment is a perturbation supported on $\{i\} \subseteq M_i$ with sup-norm $\leq \epsilon$, and every intermediate point stays in $[0,1]^d$. Worst-case invariance leaves f_θ unchanged at each increment, so resetting coordinate i from x_i to x'_i does not change f_θ. After d coordinates the point equals x', hence $f_\theta(x) = f_\theta(x')$. As x, x' were arbitrary, f_θ is constant. The number of steps, $\sum_i \lceil |x'_i - x_i|/\epsilon \rceil$, is finite for every $\epsilon > 0$. □

Corollary 5 (No-free-lunch for stage-2 patch defense)**.** *For* any *budget* $\epsilon > 0$*, no fine-tune of a non-constant classifier on* $[0,1]^d$ *can be worst-case invariant on a coordinate-covering delivery family (e.g. all* 32×32 *patches, with or without* 16*-pixel border frames) while solving a balanced* K*-class task strictly above uniform chance. In particular the impossibility holds at the small budgets* $\epsilon \approx 8/255$ *the audits use, not only in a large-budget limit.*

What this says about the open problem. Worst-case multi-surface robustness against a coordinate-covering delivery family is functionally equivalent to input-invariance, which forces the classifier to be constant *at every budget*—the chaining argument needs only that each coordinate be reachable, so the obstruction is present already at the $\epsilon \approx 8/255$ the audits use, not merely in a large-budget limit. The empirical failure of *mixed-surface stage-2* is consistent with this obstruction, but the theorem is deliberately narrower than the practical defense problem: it rules out exact global worst-case invariance, not average-case robustness on the data manifold or under sampled delivery masks. The constructive consequence is that "local-delivery transfer" should be retargeted from worst-case invariance to *distributional* delivery robustness, where the defense optimizes over sampled mask/content priors instead of requiring invariance to every submask-supported perturbation. This paper treats such distributional local-delivery robustness as the appropriate open target, because it is not foreclosed by Theorem 3.

3.6 Theorem 4: a smoothed certified-recovery objective

The proposed objective (15) is a targeted-margin adversarial unlearning objective with inner-loop attack $\delta^*(x)$. The certified counterpart replaces the inner loop with Gaussian smoothing of the margin penalty; the post-training theorem below certifies the associated binary forgotten-class detector through the Cohen–Rosenfeld–Kolter framework.

Smoothed objective. Let $\phi(z) = \max\{0, z\}$ and fix $\sigma > 0$. Define the Gaussian-smoothed margin

$$\tilde{m}_t^\sigma(x;\theta) := \mathbb{E}_{\delta \sim N(0,\sigma^2 I)}\big[m_t(x+\delta;\theta)\big].$$

$$\mathcal{L}_\sigma(\theta) \;:=\; \mathbb{E}_{x\sim D_R}\big[L_{\mathrm{ret}}(x;\theta)\big] \;+\; \lambda_f\, \mathbb{E}_{x\sim D_F}\big[L_{\mathrm{forg}}(x;\theta)\big] \;+\; \lambda_m\, \mathbb{E}_{x\sim D_F}\,\phi\big(\tilde{m}_t^\sigma(x;\theta) + \gamma\big). \tag{13}$$

Let θ^*_σ minimize (13), and define the *smoothed margin* as $\tilde{m}_t^\sigma(x;\theta)$. This is the objective implemented in `SmoothedMarginUnlearning`: the Monte-Carlo margins are averaged first and the hinge is applied to the smoothed margin. The certificate below is a property of the binary Gaussian-smoothed forget detector; if an implementation clips noisy inputs to the image cube during training or auditing, that clipped distribution is an implementation diagnostic rather than the exact Cohen–Rosenfeld–Kolter Gaussian certificate.

Theorem 4 (Certified recovery radius from smoothing)**.** *Fix* $x \in \mathcal{X}$ *and let*

$$q_t(x;\theta) \;:=\; \Pr_{\eta\sim N(0,\sigma^2 I)}\big[m_t(x+\eta;\theta) > 0\big]$$

be the probability that the forgotten class is the base model's top *logit under noise (the event $m_t > 0$). Define the* binary smoothed forget-detector $g_t(x) := \mathbf{1}\{q_t(x;\theta) > 1/2\}$*, which reports the forgotten class only when the event "the forgotten class is the top logit" is the majority outcome of the noisy base model. Let $\widehat{u}_t^{(m)}(x;\alpha)$ be a one-sided Clopper–Pearson upper confidence bound for $q_t(x;\theta)$ from m Monte-Carlo draws. If $\widehat{u}_t^{(m)}(x;\alpha) \le \tau < 1/2$, then with probability at least $1-\alpha$ over the Monte-Carlo draw we have $q_t(x;\theta) \le \tau$, and for every $x' \in \mathbb{R}^d$ with $\|x' - x\|_2 \le R$ where*

$$R \;=\; \sigma \cdot \Phi^{-1}(1-\tau), \tag{14}$$

we still have $q_t(x';\theta) < 1/2$, hence $g_t(x') = 0$: no L_2 perturbation of size $\le R$ makes the forgotten-class-top-logit event the majority outcome of the noisy base model. Consequently the binary forget-detector's recovery radius satisfies $\bar{r}_t(x;\theta) \ge R$.

Proof. The Clopper–Pearson bound gives $q_t(x;\theta) \le \tau$ with probability $1-\alpha$ over the Monte-Carlo samples. Conditional on that event, apply [13], Theorem 1, to the binary smoothed classifier over the two outcomes {forgotten class is the top logit, it is not}. The majority outcome at x is "not the top logit" with probability at least $1-\tau$, while the other outcome has probability at most τ. The binary Cohen radius is therefore

$$\frac{\sigma}{2}\big(\Phi^{-1}(1-\tau) - \Phi^{-1}(\tau)\big) = \sigma\,\Phi^{-1}(1-\tau),$$

which proves the claim. □

Remark (binary vs. multiclass). The certified event is binary—*the forgotten-class-top-logit event is not the majority outcome under noise*—which is exactly the security-relevant condition (an attacker recovering the forgotten label). It is *not* a statement about the full multiclass smoothed argmax: $q_t < 1/2$ leaves open which non-forget class is predicted, and the forgotten class could still be the largest single multiclass probability below 1/2. The theorem guarantees only that the forgotten-class-top-logit event is not the majority outcome of the binary detector. A multiclass-argmax certificate would additionally require the usual top-vs-runner-up smoothing gap.

Corollary 6 (Generalization of the certified-recovery fraction)**.** *Fix a target level $\tau < 1/2$ and let $R = \sigma\,\Phi^{-1}(1-\tau)$ be the corresponding certified radius from Theorem 4. For a fixed trained model and fixed threshold, define the empirical certification indicator*

$$\widehat{C}_\tau(x) := \mathbf{1}\{\widehat{u}_t^{(m)}(x;\alpha_i) \le \tau\}.$$

Let $Q_\tau := \Pr_{x\sim\mu_t}[\widehat{C}_\tau(x) = 1]$ be the population fraction of forget inputs that the Monte-Carlo audit would certify under the same procedure, and let $\widehat{Q}_\tau$ be its empirical value on n_f i.i.d. forget samples. Then Hoeffding's inequality gives, with probability at least $1-\delta$ over the sampled inputs,

$$Q_\tau \;\ge\; \widehat{Q}_\tau \;-\; \sqrt{\tfrac{\log(1/\delta)}{2n_f}}.$$

Per-input Monte-Carlo validity is separate: each certified point has probability at least $1-\alpha_i$ of satisfying the true detector guarantee in Theorem 4. For simultaneous validity over the n_f audited inputs, set $\alpha_i = \alpha/n_f$ and union bound; otherwise the guarantee is per-input.

Why this is the missing piece. The objective (15) is an *empirical* adversarial unlearning objective in the AMUN [25] mold: minimize a worst-case inner attack. It carries no certificate. Prior work already connects randomized smoothing to certified unlearning [14], but it certifies the stability of *quantized gradients* under data removal—a parameter-space guarantee about the unlearning *operation*—rather than a guarantee about model behavior on perturbed inputs. To my knowledge, Eq. (13) is an early unlearning objective that yields a per-input *input-space* randomized-smoothing lower bound on forgotten-class recovery radius, and Corollary 6 transports the fixed-threshold certification fraction to the forget distribution with a standard binomial/Hoeffding interval. The existing recovery-radius audit provides the matched empirical companion: predicted-vs-realized radii are a consistency diagnostic for (13)'s training (an L_2 diagnostic; the L_∞ audit relates to it only up to the $\sqrt{d}$ conversion).

3.7 Empirical claims, restated as theorem diagnostics

The four theorems above re-cast the benchmark results of this paper, and are checked or diagnosed empirically in Section 8:

1. **Recovery–certification scatter (Thm 1).** The cross-method $(\hat{L}, \mathrm{median}(\hat{r}))$ scatter plot ranks methods under (5): a larger $\hat{L}$ at comparable margin forces a smaller certifiable radius. Drawing the bound line $\hat{r} = -\bar{m}/\hat{L}$ itself would need the per-method oracle margin $\bar{m}$, which an oracle-free auditor lacks, so the figure is a ranking diagnostic rather than a tightness test.

2. **Selective vs. generic (Thm 2).** The forget–retain $\hat{p}_F, \hat{p}_R$ table per method becomes a Pinsker-bounded KL comparison; large rate gaps force a quantitative KL lower bound, while near-zero rate gaps do not certify absence of leakage. The larger-n oracle-null audit is the stronger empirical calibration: SalUn and the smoothed-margin row separate from the retain-only null, while SCRUB's signed selectivity is *unstable across seeds* (selective on 42, significantly anti-selective on 123, at parity on 456; bootstrap CIs in Section 8), so no stable selective or generic-fragile label survives a single-seed reading.

3. **Local-delivery negative (Thm 3).** The theorem rules out exact worst-case invariance over a coordinate-covering delivery family. The mixed-surface stage-2 failure is consistent with that obstruction, but average-case transfer under realistic sampled mask distributions remains an open defense target.

4. **Certified objective (Thm 4).** The smoothed objective (13) replaces the inner attack in the empirical objective (15). Trained end-to-end ($\sigma = 0.10$) across seeds 42, 123, and 456, it yields a post-training L_2 certificate for the binary smoothed forgotten-class detector. The native L_2 deterministic recovery audit is a companion diagnostic whose measured medians sit above the detector-radius scale; the converted L_∞ plot is only a norm-mismatch caveat.

3.8 Relation to prior inevitability, impossibility, and certification results

These results position the paper relative to recent impossibility and certification work:

- RURK [6] establishes high-dimensional *inevitability* of residual knowledge in expectation; Theorem 1 converts inevitability into a *quantitative budget* the audit can estimate without an oracle.

- [24] establishes path-dependence impossibility for retrain equivalence in multi-stage training; Theorem 3 establishes a *different* impossibility — multi-surface worst-case invariance — for the same end-goal (distinguishability), but with a coverage/connectivity rather than path-dependent mechanism.

- AMUN [25] uses adversarial examples as an *unlearning* primitive; Theorem 4 inverts the role and uses smoothing as a *certified-defense* primitive against post-hoc recovery attacks.

The benchmark is no longer the headline; it becomes the experimental validation of a small theory.

3.9 A controlled probe of the fragility scale

Proposition 1 reads the elasticity b off an *observational* cross-checkpoint regression in which the Lipschitz scale L co-varies with the unlearning method. A within-method controlled estimate would dial a single robustness knob, holding the method fixed, and identify b causally. We attempted this and report it as an honest negative, because the negative result is itself informative about what governs the recovery scale.

Weight-decay instrument: a first-stage null. Holding the method (SalUn), the base, and the forget class fixed, we varied the LoRA adapter's weight decay across $\{0, 10^{-3}, 10^{-2}, 10^{-1}\}$ and re-audited. The retain-control radius r_R did not move: across four orders of magnitude of regularization, Spearman(wd, r_R) $= -0.07$ and r_R stayed within audit noise of 6.0×10^{-3} (the lone exception being a single seed at wd $= 0$). With no first stage there is no identified elasticity; the instrument is invalid. Mechanistically, the recovery radius is governed by the input-space geometry of the *base* representation ($\sim$95% of parameters, frozen), and a low-rank adapter's weight norm is too weak a lever to move it.

The scale is architecture-level, not init-level. A variance decomposition of $\log r_R$ over the 132-audit set (one architecture, `vit_tiny`; three independently trained base seeds; twelve method families) corroborates this. The base seed explains essentially none of the variance ($\eta^2_{\text{seed}} = 0.002$; the three seed means of $\log r_R$ agree to within 0.06), whereas the unlearning method explains $\eta^2_{\text{method}} = 0.25$ and within-method regularization explains ≈ 0 (the weight-decay null above). The global fragility scale is thus a property of the *architecture*: invariant to the random initialization of the base, invariant to adapter-side regularization, and only modestly reshaped by the unlearning objective. This is the cleanest available statement of where the recovery radius "lives," and it points to the right instrument: the knob that moves r_R is the base *geometry*, so the controlled experiment must vary base smoothness while holding the unlearning method fixed.

Base-smoothness instrument: a clean causal estimate. We realize this with adversarial pre-training strength. Using five ViT-tiny CIFAR-10 bases spanning a range of PGD-training strengths (from a clean base to strongly robust ones) and unlearning each identically with SalUn, the base-smoothness knob moves the retain-control radius over a 22× range ($r_R \in [5.7 \times 10^{-3},\ 1.4 \times 10^{-1}]$)—a decisive first stage, in contrast to the weight-decay null. Crucially this instrument is *clean*: adversarial training reshapes the global input-space geometry that sets r_R without a separate forget-suppression channel (unlike the forget/retain tradeoff weight, which moves r_R but directly controls forget margins and so cannot identify the fragility elasticity). Regressing $\log r_F$ on $\log r_R$ across the ten checkpoints (five strengths × two seeds) gives a within-method causal slope

$$\hat{b}_{\text{causal}} = 1.36 \quad \text{(cluster-bootstrap 95\% CI [1.31, 1.84]),}$$

excluding the single-scale null $b = 1$ and statistically indistinguishable from the observational cross-method estimate $\hat{b} = 1.31$ of Section 8 (Figure 1D). The super-elasticity of Proposition 1 therefore survives a controlled test: it is not an artifact of pooling methods of differing fragility, but a within-method causal response of the forget radius to the base smoothness scale. The audit budget must be widened for the robust bases (their radii exceed the default 8/255 and are otherwise right-censored); we audit at $\epsilon_{\max} = 0.25$ so all ten medians are measured at success rate ≥ 0.92.

Cross-architecture replication. The causal estimate above is within a single architecture and dataset (`vit_tiny`, CIFAR-10). To test whether the mechanism is architecture-bound we re-run the *observational* cross-method fit on two new architecture/dataset pairs—ResNet-18 (a convolutional network) and ViT-Small, both on CIFAR-100—with six LoRA unlearning methods each (SalUn, SCRUB, CE-ascent, Uniform-KL, feature-scrub, smoothed-margin) under the identical recovery-radius protocol. Pooling the per-method points ($n = 11$; one SalUn/ViT-Small point is dropped because its forget radius collapses to exactly 0, undefined on a log axis and itself direction-consistent) reproduces both halves of the mechanism: the elasticity is $\hat{b} = 1.98$ (case-resample 95% CI [1.34, 3.13], with $b > 1$ in every leave-one-out fit, range [1.73, 2.26]) and the selectivity–fragility confound is $r = -0.74$ (CI $[-0.92, -0.47]$, $r < 0$ in every leave-one-out fit, range $[-0.80, -0.68]$). This is an observational fit at a single seed per cell, so it pins the *direction*—super-elasticity ($b > 1$) and a negative confound on a CNN and a second ViT over a new dataset—rather than the exponent, which here (≈ 2) exceeds both the CIFAR-10 benchmark (1.31) and the within-method causal estimate (1.36); a multi-seed cross-architecture *causal* instrument is the remaining step. Numbers in `results/analysis/metrics/crossarch_pooled.json` (script `scripts/analysis/51_crossarch_pooled.py`).

4 Related Work

These experiments sit at the intersection of machine unlearning, adversarial robustness, and parameter-efficient adaptation. The benchmark builds on SalUn [2], SCRUB [1], LoRA [3], VPT [5], and AutoAttack [4], and treats recent 2025–2026 methods such as RURK [6] as newer literature baselines; it also includes a gradient-ascent row labeled CE-U, which implements cross-entropy *ascent* rather than the corrected-target objective of [11] and should be read as a GA baseline (see Appendix I). It additionally includes adapted recent objectives such as SGA [8] and BalDRO [9] as cross-domain benchmark rows. ORBIT is introduced here as a diagnostic stress-test objective rather than as an external literature anchor. Additional appendix-only proxy and supporting rows, including FaLW [12] and Langevin-style [10] variants, are used only when they help define regime boundaries. Langevin Unlearning [10] is used as a theoretical anchor for principled privacy-certified evaluation, and MUSE [7] motivates multi-regime testing rather than single-metric forgetting claims.

A wave of concurrent 2025–2026 work studies the same residual-knowledge phenomenon from complementary angles. Xuan and Li [15] verify robust unlearning with the Unlearning Mapping Attack, an adversarial-query probe for forgotten traces; recovery-radius auditing quantifies the same probe as a per-example minimum perturbation and ties it to a certificate (Theorems 1 and 4) and a forget-vs-retain selective test (Theorem 2). Goel et al. [16] audit unlearning with an information-theoretic Partial Information Decomposition, and Yu et al. [18] certify visual unlearning at the *representation* level (linear-probe recovery, CKA, feature separability), both finding that output-level forgetting hides residual structure; Theorem 2 is a coarser but distribution-free counterpart that bounds the KL of the binary recoverability outcome rather than decomposing mutual information, and the recovery radius is a complementary *input-space* quantity to Mirage's feature-space audit.

Method / Row	Fidelity	Paper Role	Current Read
SalUn (faithful full-ft)	faithful	main text	core literature anchor
SCRUB distill	reasonably faithful	main text	literature baseline; hard negative / generic fragility
ORBIT	diagnostic baseline	main text	stress-test objective introduced here; best later defense row
RURK	RURK-*style* (random noise)	main text	robustness term uses random, not adversarial, perturbations
SGA	label-smoothed GA (faithful core)	auxiliary	generic-fragility cross-domain benchmark row
BalDRO	BalDRO-*style* (loss reweighting)	main text	softmax hard-sample reweighting, not min–max DRO
CE-ascent (labeled CE-U)	GA baseline, *not* CE-U	main text	implements gradient ascent, not the corrected-target CE-U objective
robust-base SalUn	faithful frontier	main text	utility-vs-recovery frontier
Uniform KL	bridge / supporting	appendix	seed-sensitive bridge target
Prune+KL	supporting	appendix	auxiliary clean-forgetting row
CNN cross-checks	supporting	appendix	architecture sanity checks
legacy manifold/imposter	supporting	appendix	older harder-than-baseline branch
LoRA SalUn	proxy	appendix	PEFT-side approximation only
OrthoReg / noisy_retain	proxy	appendix	repo-side approximations only

Table 1: Method classification used in the paper. The table separates literature anchors, appendix-only supporting rows, and proxy artifacts so that the main benchmark reflects only rows with defensible evaluation fidelity.

The closest hypothesis-testing framing, Pandey and Kulkarni [23], tests which samples to edit out of a distribution rather than whether a forgotten example is selectively recoverable, so it is orthogonal to the forget-vs-retain test here. The qualitative *distinction* between residual-knowledge leakage and generic adversarial fragility is not new—RURK [6] and the UMA probe of Xuan and Li [15] both raise it—but these treat it as a binary caveat and, as Xuan and Li note for their own attack, do not measure baseline robustness, correlate it with apparent leakage, or calibrate against a retrained oracle. The contribution here is to make that confound *quantitative*: a benchmark-wide negative correlation between selectivity and global fragility (Section 8), a super-linear forget-radius elasticity that explains it through Theorem 1, and a fragility-matched taxonomy whose validity is checked by the oracle null landing where it must. Ha et al. [17] introduce a prototypical relearning attack that restores forget-set performance from per-class prototypes with only a few samples; that is a weight- and prototype-space relearning attack rather than the input-space recovery studied here, and it is precisely the channel ORBIT's dispersion penalty is designed to weaken.

This paper's certification angle is positioned against the certified-unlearning line specifically. Langevin unlearning [10], Rewind-to-Delete [20], the neural-network certificates of [21], and retain-sensitivity calibration [19] certify a *parameter-distribution* (ϵ, δ): indistinguishability between the distribution over unlearned weights and retraining from scratch. As a recent verification survey [22] notes, these guarantees say nothing about whether a single deployed model resurfaces the forgotten class under a crafted input. The recovery radius is the natural measure of that uncertified input-space axis, and Theorem 1 makes the separation precise (Section 3): the paper's title names a gap that is structural, not rhetorical.

5 Experimental Setup

5.1 Models and data

The main benchmark uses CIFAR-10 with 224×224 inputs and ViT-tiny-based checkpoints; Section 8 adds single-seed CIFAR-100 cross-checks. The main rows are SalUn and SCRUB as literature anchors, RURK and CE-U as newer literature methods, SGA and BalDRO as adapted recent methods, ORBIT as a diagnostic stress-test objective introduced here, and robust-base SalUn as a defense frontier. Several appendix-only proxy, bridge, and negative-control rows are retained only as supporting comparisons.

5.2 ORBIT: diagnostic stress-test objective

ORBIT (oracle-aligned, representation-dispersive unlearning) is a diagnostic objective introduced here to stress-test recovery-radius audits, not an external literature baseline. Its objective combines four terms: (1) a hinge-margin loss suppressing the forget-class logit below the maximum non-forget logit; (2) masked dark-knowledge distillation from a frozen oracle on non-forget classes, with the forget dimension excluded from the KL target; (3) a feature alignment term pulling forget-input representations toward the oracle's feature space; and (4) a dispersion penalty pushing forget features apart to weaken prototype and linear recovery channels. The motivation is that suppressing a single output logit is insufficient when the underlying representation still encodes the forgotten concept—alignment and dispersion together target the representational substrate rather than only the output layer. ORBIT is the strongest patch-positive case in the benchmark, making it the primary vehicle for studying local-delivery attacks and patch-aware defenses.

5.3 LoRA as benchmark substrate

The repository originated as a LoRA-centered audit, and many of the stored checkpoints remain LoRA adapters for comparability and fast stage-2 iteration. That implementation choice is useful experimentally, but it is not the main scientific claim of the paper. The paper is about unlearning robustness under multiple attack regimes. LoRA is therefore treated primarily as part of the benchmark substrate, while selected full-finetune and cross-architecture rows check that the resulting story is not purely an artifact of one parameter-efficient adaptation scheme.

5.4 Baseline checkpoint quality

The clean baseline summary shows that unlearning methods can appear successful on clean metrics while still exposing recovery channels. Across seeds 42, 123, and 456, clean retain accuracy stays in the low 90s for the main baselines, while forget accuracy varies dramatically by method. The base model sits at 91.83 ± 4.36 forget and 91.80 ± 3.77 retain. Uniform KL, CE Ascent, SalUn, and SCRUB all preserve similar retain performance, but their forget-side variance is extremely large, which already suggests that clean-point forgetting is an unstable proxy for actual robustness. These clean baselines therefore motivate attack-based evaluation rather than one-number forgetting claims.

5.5 Recovery metric

For each forget or retain-control sample, the audit searches for the minimum perturbation that causes the unlearned model to predict the forgotten class. Clean forgotten-class predictions are assigned radius 0. If the optimizer succeeds within the audit budget $\epsilon_{\max}$, binary search returns a finite estimate $\hat{r}_t(x) \leq \epsilon_{\max}$; if it fails, the sample is right-censored as $\hat{r}_t(x) > \epsilon_{\max}$. To keep the empirical object aligned with the theoretical distribution of r_t, rows should be read through three statistics: recovery rate at $\epsilon_{\max}$, the censored all-sample median (failures counted as right-censored above $\epsilon_{\max}$, so the median may be reported as $> \epsilon_{\max}$), and the success-conditioned median among finite successful recoveries. Several compact tables report the success-conditioned median for space; "undefined" means no finite recoveries, not a radius of zero. The interpretive question is whether forget examples are selectively easier to recover than matched retain controls under the same censoring rule. A consistent forget-vs-retain gap supports the view that the attack is exposing residual information left behind by incomplete unlearning, whereas non-selective success on both groups is more consistent with generic adversarial fragility.

Regime	Threat Model	Primary Metric	What Counts as a Defense Win
Conditional recovery	per-example full-image targeted recovery	forget median / retain median / clean-success	higher forget median without comparable retain-side degradation
Conditional patch	local one-patch delivery	forget success / retain-to-forget success / alpha	lower forget success without added retain-to-forget failure
Conditional frame	border-frame local delivery	forget success / retain-to-forget success / alpha	lower forget success without transfer failure on retain
Conditional multi-patch	multi-local delivery	forget success / retain-to-forget success / alpha	lower forget success under richer local geometry
Amortized short-budget	learned low-step attack	forget success / retain success / median radius	stronger selective low-step recovery than plain low-step optimizer
Robust-base frontier	utility-recovery tradeoff	retain accuracy vs recovery radius	better operating point, not a binary robust point

Table 2: Protocol table for the benchmark. The benchmark is designed as a controlled robustness microscope: each regime tests a distinct failure mode, and a defense is only treated as positive when it improves the regime-relevant robustness metric without introducing comparable retain-side failure.

6 Multi-Regime Results

This work evaluates machine-unlearning robustness under four primary recovery-radius regimes: strong per-example conditional recovery, conditional patches, border-frame delivery, and multi-patch local delivery. Across the main benchmark rows audited in depth—SalUn, SCRUB, RURK, CE-U, BalDRO, and ORBIT—the dominant threat remains targeted per-example recovery, while universal perturbations are mostly ineffective on the stronger checkpoints. The point of this matched audit is not to ask whether a target-class adversarial perturbation exists in the abstract. It is to ask whether forget examples are selectively easier to drive back to the forgotten class than matched retain controls. Figure 2 and Table 3 summarize the main defense result: feature-subspace stage-2 gives positive point-estimate shifts on BalDRO, SalUn, and ORBIT without collapsing clean accuracy, with the clearest practical signal on ORBIT and SalUn and only a tiny BalDRO shift. Figure 3 shows that robust-base SalUn does not eliminate leakage outright; instead it moves along a utility-vs-recovery frontier. Figure 4 shows that conditional patches remain viable, but only for a subset of methods.

Feature-subspace stage-2 defense. The most promising post-hoc defense family in the current benchmark is a feature-subspace stage-2 finetune. Rather than modifying the entire forgetting objective from epoch 0, the defense begins from an already-unlearned adapter and suppresses attacked forget-sample features inside a learned recovery subspace. This regime avoids the instability of earlier logit-margin defenses and produces positive point-estimate shifts on BalDRO, SalUn, and ORBIT. The effect is clearest on ORBIT and SalUn; the BalDRO shift is numerically tiny, and the table does not yet include the paired confidence intervals needed to treat it as statistically established. The effect is modest in absolute magnitude because the benchmark is CIFAR-10 under an L_∞ budget on 224×224 inputs, where successful radii are numerically small by construction. What matters in the current evidence is that some rows shift the forget-vs-retain frontier in the right direction without sacrificing clean retain utility, not that feature-subspace training is a universal defense.

CE-U as a core baseline. CE-U belongs in the main benchmark even though its first qualified positive defense row appears in the patch regime rather than under full-image recovery. Under the 2-step pilot audit (n=8, seed 42), the baseline CE-U checkpoint has forget median 0.00368 with 0% clean forget success, and retain-control attacks succeed 0 of 8 times within the ϵ budget. This is suggestive pilot evidence for selectivity, but it is not ranked against SalUn or ORBIT because those rows use larger full audits. CE-U full-image feature-subspace stage-2 variants are at best flat, and the combined row is slightly worse under the APGD cross-check, so the CE-U defense claim is reserved for the qualified combined stage-2 patch result discussed below.

Model	Setup	Forget Median*	Retain Median*	Clean Forget Rate
BalDRO[§]	baseline	0.00343	0.00846	0.000
BalDRO[§]	feature-subspace stage-2	0.00349	0.00858	0.000
SalUn	baseline	0.00251	0.00564	0.031
SalUn	feature-subspace stage-2	0.00276	0.00600	0.031
ORBIT	baseline	0.00340	0.00643	0.031
ORBIT	feature-subspace stage-2	0.00423	0.00690	0.000
RURK[§]	baseline	0.00159	0.00576	0.062
RURK[§]	feature-subspace stage-2	0.00159	0.00576	0.062
CE-U[†§]	baseline	0.00368	—[‡]	0.000
SCRUB	baseline only	0.01078	0.00980	0.000

*Finite-success medians, not censored all-sample medians; recovery rates and censoring are tracked in the JSON artifacts.

[†]CE-U: 2-step pilot audit, n=8, seed 42.

[‡]0 of 8 retain-control attacks succeeded within ϵ; finite-success retain median undefined.

[§]Objective-level adaptation, not a faithful reproduction: BalDRO-*style* softmax reweighting, RURK-*style* random-noise robustness, and CE-*ascent* (labeled CE-U). See Table 1 and Appendix I.

Table 3: Finite-success recovery-radius medians under the strongest per-example conditional attack. Feature-subspace stage-2 gives positive point-estimate shifts for BalDRO, SalUn, and ORBIT, but this table should not be read as a significance table: BalDRO's shift is tiny, and the available stage-2 rows are not fully seed-paired against every baseline. RURK and CE-U are flat under this full-image defense family; SCRUB is a seed-sensitive negative control with nearly identical forget and retain radii on most seeds. CE-U retains 0 of 8 retain-control attacks in the pilot audit, which is suggestive but not statistically comparable to the full-audit rows. Rows use $n = 32$ samples per class per seed where available, except where noted. The distributed SalUn baseline checkpoint does not reproduce its reported row; regenerated 3-seed SalUn baseline numbers are in the reproducibility appendix.

Robust-base frontier. Robust-base SalUn does not behave like a binary defense success. Instead, it exposes a frontier: at low utility, conditional recovery can drop to zero on the sampled audit; as retain accuracy rises, recovery reappears. This is a stronger framing than a single checkpoint claim because it shows that robustness and post-unlearning utility are coupled. Robust-base training is therefore treated as an operating-point choice rather than a one-shot fix.

Conditional patches. The patch result is not a universal trigger. Universal static patches were mostly ineffective on the stronger methods. The viable regime is instead a conditional small-area patch attack whose content and placement depend on the input. ORBIT is the flagship positive case: a 32×32 patch covering 2.04% of the image recovers forgotten behavior strongly while inducing no retain-to-forget hijack. SalUn is patch-vulnerable only at a larger 48×48 budget, and BalDRO is effectively negative at both tested patch sizes. This method dependence reinforces the broader paper thesis: residual forgetting leakage is structured, but not universal.

Delivery surfaces beyond one patch. The later delivery-surface checks strengthen the attack story and narrow the defense story. Conditional frame attacks and conditional multi-patch attacks are both real local delivery surfaces, but they do not behave like simple extensions of the one-patch regime. Patch-aware defenses that help on one square patch often fail to transfer: the ORBIT patch-stage2 row becomes easier to recover under the frame attack, CE-U combined stage-2 also becomes easier and adds retain-side failure under the frame attack, and faithful full-ft SalUn patch-stage2 becomes easier under multi-patch delivery. The main exception is weak positive transfer from the ORBIT patch-stage2 row to multi-patch delivery. Frame and multi-patch regimes are therefore treated as first-class red-team surfaces rather than appendix-only perturbation variants.

Second-wave defenses. Two later branches sharpen the story. First, teacher-guided stochastic CVaR provides a small but reproducible full-image defense improvement on ORBIT across two seeds, whereas the same probabilistic family remains neutral on CE-U. Second, a combined subspace-plus-patch stage-2 defense gives the first CE-U row that improves under matched Adam audits while also reducing conditional patch success, although its APGD result is still slightly worse than baseline. By contrast, the larger mixed-surface stage-2 defense was explicitly designed to transfer across one-patch, frame, and multi-patch delivery and failed to do so reliably. These later rows are therefore useful but qualified: they extend the defense story beyond pure feature-subspace training without becoming universal fixes.

Takeaway. The combined evidence supports a multi-regime view of red-teaming machine unlearning. Strong per-example conditional attacks remain the main threat, conditional patches expose an additional local attack surface on some methods, and frame plus multi-patch attacks show that local-delivery robustness is materially harder than one-patch robustness alone. The best defense families found here are modest and method-specific: feature-subspace stage-2 gives the clearest point-estimate gains on ORBIT and SalUn, teacher-guided stochastic CVaR helps ORBIT slightly, and CE-U combined stage-2 is only a qualified partial row. RURK remains a hard case, and SCRUB serves as a seed-sensitive negative control where generic adversarial fragility usually overwhelms forget-specific interpretation.

7 Short-Budget and Auxiliary Regimes

7.1 Amortized short-budget attack

The learned distilled short-budget attacker is target-dependent. It improves matched low-step Adam on ORBIT across both seeds and also helps on SalUn, but it does not preserve a clean forget/retain Pareto on RURK. It is therefore positioned as an amortized attack regime rather than a universal attack method.

7.2 Earlier prompt/probe regime

An earlier prompt-style red-team branch found a strong effect on Uniform KL under an oracle-contrastive probe, but not on the stronger methods. In particular, the strongest earlier prompt result moved Uniform KL seed 42 from 12.70% forget to 21.10% forget with only a 0.36pp retain drop, while SalUn, SCRUB, and ORBIT remained in the resistant bucket under that family. This is supporting evidence for regime sensitivity rather than the core attack result.

7.3 Uniform KL as a bridge target

Uniform KL also remains useful under the newer conditional-recovery regime. A fast matched pilot with 2-step `adam_margin` on seed 42 recovers all sampled forget examples with median radius 0.00147 while succeeding on only one of eight sampled retain controls. The same pilot on seed 123 is completely flat. This makes Uniform KL a bridge target between the older probe story and the newer conditional-optimization story: it is not only probe-vulnerable, but also locally recoverable

Target	Seed	Attack	Forget Succ.	Retain Succ.	Forget Median
ORBIT	42	2-step Adam	9.4%	0.0%	0.02500
ORBIT	42	Distilled + 2-step refine	18.8%	0.0%	0.02083
ORBIT	123	2-step Adam	18.8%	0.0%	0.00316
ORBIT	123	Distilled + 2-step refine	31.2%	3.1%	0.01189
SalUn	42	2-step Adam	71.9%	9.4%	0.00435
SalUn	42	Distilled + 2-step refine	84.4%	9.4%	0.00447
SalUn	123	2-step Adam	68.8%	6.2%	0.00328
SalUn	123	Calibrated distilled + 2-step refine	81.2%	12.5%	0.00389
RURK	42	2-step Adam	87.5%	6.2%	0.00175
RURK	42	Calibrated distilled + 2-step refine	90.6%	18.8%	0.00175

Table 4: Short-budget comparison between matched 2-step Adam and the learned distilled conditional attacker. The learned attacker helps most on ORBIT, helps on SalUn with a weaker forget/retain Pareto, and does not remain selective on RURK. All rows report metrics over $n = 32$ audit samples per class.

Target	Seed	Regime	Forget Succ.	Retain Succ.	Forget Median / Alpha
Uniform KL	42	2-step conditional pilot	100.0%	12.5%	0.00147
Uniform KL	123	2-step conditional pilot	0.0%	0.0%	N/A
Uniform KL	42	32×32 conditional patch	15.6%	0.0%	0.15234
Uniform KL	42	feature-subspace stage-2 pilot	100.0%	12.5%	0.00147
SGA	42	full conditional audit	100.0%	100.0%	0.00294
SGA	123	full conditional audit	96.9%	100.0%	0.00968
SGA	456	full conditional audit	96.9%	96.9%	0.01091
SGA	42	32×32 conditional patch	0.0%	0.0%	N/A
SGA	42	2-step baseline pilot	75.0%	0.0%	0.00417
SGA	42	2-step stage-2 pilot	62.5%	0.0%	0.00441

Table 5: Auxiliary seed-sensitive targets. Uniform KL is a bridge case between the older probe regime and the newer conditional-recovery regime, while SGA behaves like generic adversarial fragility under the full audit, stays patch-negative, and only shows a weak fast-pilot stage-2 effect. Audits use $n = 32$ samples per class unless otherwise specified as a pilot ($n = 8$).

on the vulnerable seed. A conditional 32×32 patch on seed 42 also yields a weak but nontrivial patch result (15.6% forget recovery and 0% retain-to-forget success), which is enough to show that KL is patch-accessible without making it a flagship patch case.

7.4 Additional auxiliary targets

SGA is another useful auxiliary target from the recent related-work list, but the full conditional audit does not support a selective-leakage reading. Across seeds 42, 123, and 456, forget and retain-control success rates are both near ceiling, so this row looks like generic adversarial fragility rather than a meaningful forget-vs-retain gap. A matched 32×32 conditional patch is negative on seed 42, and a lightweight feature-subspace stage-2 pilot yields only marginal improvement. Uniform KL and SGA are therefore auxiliary regime-defining targets rather than the main methods around which to build the defense story.

8 Empirical Validation of the Theory

Each of the four theorems in Section 3 comes with a benchmark check. This section reports those checks and connects the attack regimes to the theory rather than treating them as standalone attack curves.

8.1 Theorem 1: the recovery–certification scatter

For each method, the margin-Jacobian audit estimates a local Lipschitz proxy $\hat{L}$, and the recovery-radius audit estimates the median radius median($\hat{r}$). The underlying join contains 40 (method, seed, config) rows; Figure 5 shows method-level aggregates to make the ranking readable. It is a diagnostic for the ordering suggested by Eq. (5), not a tightness plot: drawing the actual bound $\hat{r} \geq -\bar{m}/\hat{L}$ would require the per-method oracle margin $\bar{m}$, and the plotted $\hat{L}$ is a local logit-margin proxy rather than the probability-margin constant in Corollary 2. The aggregates separate the methods cleanly ($\hat{L} \approx 47$ for SCRUB, 26 for ORBIT, 16 for Uniform-KL, and 15 for SalUn), so this section uses the scatter as a heuristic ranking and sanity check, not as a measured certificate. The empirical recovery rates that feed this scatter are themselves attack-valid in one direction without assuming optimizer completeness: by Corollary 1 each successful attack at radius $\leq \rho$ certifies $r_t \leq \rho$ for that input, so the measured rate lower-bounds $\Pr_{\mu_t}[r_t \leq \rho]$ while Theorem 1 upper-bounds it—a two-sided sandwich that does not rest on the inner attacker being complete.

A per-sample validation, not just four aggregates. The aggregate scatter is intentionally coarse, so we also test Eq. (5) at the *sample* level, where it makes a sharp prediction: since $r_t \gtrsim -m_t/L$, individual samples with a less-suppressed forget logit should recover at smaller radius. Using $\log p_t(x)$—the log clean forgotten-class probability, monotone in the logit margin (the probability margin saturates near -1 once p_t is tiny, so the *log* carries the usable signal)—Theorem 1 predicts $\mathrm{corr}(\hat{r}, \log p_t) < 0$. Across the 58 audited checkpoints that retain genuine forget-margin variation (excluding blocks where every forget logit is uniformly crushed to underflow), the median per-checkpoint correlation is -0.64, with 97% of checkpoints negative and 83% strongly so (< -0.3); matched retain controls obey the same law (median -0.45, 86% negative), confirming this is the general margin–radius relation of Theorem 1 rather than a forget-specific artifact (`scripts/analysis/45_per_sample_thm1.py`). This is a hundreds-of-points-per-checkpoint test of the pointwise inequality, not a four-aggregate ranking, and it lets the next subsection's mechanism rest on the validated pointwise law.

8.2 Theorem 2: the Pinsker KL table

The recovery-rate audit computes forget and retain-control rates $(\hat{p}_F, \hat{p}_R)$ for 226 (method, seed, audit-config) rows and reports the Pinsker lower bound $\mathrm{KL} \geq 2|\hat{p}_F - \hat{p}_R|^2$ (and the tighter Bretagnolle–Huber bound) in the repository artifacts. The main full-audit rows show the expected qualitative ordering: SalUn has a large measured gap, ORBIT has a positive but smaller gap, and SCRUB has nearly matched rates ($\hat{p}_F \approx \hat{p}_R \approx 0.97$, $|\Delta| \approx 0.031$). Corollary 3 gives these gaps an operational reading: an observer who sees only the single recovery bit can label an input forget-vs-retain with accuracy $\frac{1}{2}(1 + |\hat{p}_F - \hat{p}_R|)$, so SalUn's gap is a sizeable above-chance advantage while SCRUB's $|\Delta| \approx 0.031$ is barely above a coin flip. The SCRUB interpretation rests on this direct measured equality, not on Pinsker: Pinsker only lower-bounds KL when a genuine gap exists and cannot certify an upper bound on leakage. CE-U is kept as pilot evidence rather than a ranked headline row: its seed-42, $n = 8$ audit has $\hat{p}_F = 0.875$ and $\hat{p}_R = 0.000$, which is suggestive of selectivity but too small to compare statistically against the full-audit rows.

A stronger control: the oracle null and larger n. The rate table above is limited in two ways the literature rightly flags: small per-row n, and the worry that a forget-vs-retain radius gap is merely the intrinsic difficulty asymmetry of driving *any* input to the forgotten class. A larger-n ($n_F = n_R = 128$) `adam_margin` audit and a *retain-only oracle null*—a model retrained on the retain set only, with the same K-class output space and the forgotten-class logit left without positive training examples—address both issues. This null is not an information-theoretic proof of zero class knowledge, especially if pretrained features are used; it is a matched operational baseline for target-label fragility. Across seeds 42, 123, and 456 the oracle null shows essentially no intrinsic asymmetry (forget/retain median-radius ratio 0.94, 1.05, 1.01), so a gap beyond the null on an *unlearned* model is evidence of residual-knowledge signal rather than baseline artifact. Against this null, SalUn separates cleanly (forget median radius 0.000 vs retain 0.0039 at n=128, non-overlapping bootstrap CIs) and the smoothed-margin model recovers well below the null—both genuine residual leakage under this audit. SCRUB, by contrast, does not merely vary in magnitude across seeds—its selectivity *changes sign*. Reporting the forget-vs-retain median-radius ratio as "× easier" (retain over forget, so > 1 is selective) with 5000-resample bootstrap 95% CIs: seed 42 is selective (2.04×, CI $[1.85, 2.38]$), seed 123 is significantly *anti*-selective—forget is *harder* to recover than the retain control (0.83×, CI $[0.74, 0.93]$, excluding parity)—and seed 456 is at parity (1.01×, CI $[0.82, 1.21]$). Both the seed-42 and seed-123 CIs exclude 1, in *opposite* directions, against a flat oracle null on every seed. So the earlier "selective on 42, parity on 123/456" reading was too generous to the parity story: SCRUB is not consistently generic-fragile, it is *unstable in the sign of its leakage*, which is a stronger reason to distrust any single-seed selective claim about it. One observation suggests where to look next, though $n = 3$ seeds cannot establish it: the lone selective seed (42) is also the one whose SCRUB checkpoint is globally most fragile (smallest retain-control median radius, 0.0052 versus 0.0092 and 0.0123), hinting that SCRUB's apparent selectivity is a property of an unstable, globally fragile checkpoint rather than of the method. The next paragraph runs exactly that test—correlating signed selectivity against the oracle-free fragility scale—not just over more SCRUB seeds but across every audited checkpoint, and finds the confound is benchmark-wide. This instability is consistent with the large clean-baseline forget variance of Appendix A and is quantified per seed in `results/analysis/metrics/scrub_seed_dependence.json`.

A benchmark-wide fragility confound in the selectivity metric. The SCRUB observation generalizes. Pooling *every* (method, seed, attack config) audit row that carries a matched forget and retain control—132 rows over 57 distinct unlearned checkpoints—let $S = \log(\tilde{r}_R/\tilde{r}_F)$ be the signed selectivity (median retain radius over median forget radius, so $S > 0$ is selective) and let the *fragility scale* $F = \tilde{r}_R$ be the retain-control median radius, an oracle-free measure of how globally easy the checkpoint is to attack. Across the benchmark S is negatively correlated with $\log F$: Pearson $r = -0.25$ (Spearman -0.37), with a suite-clustered bootstrap 95% CI of $[-0.44, -0.09]$ that excludes zero. The effect is modest—fragility explains only $\sim$6% of selectivity variance, and 83% of even the most robust checkpoints still read as nominally selective—but it is *not* an artifact of pooling: it survives restriction to CIFAR-10 alone ($r = -0.29$; clustered CI $[-0.49, -0.04]$), to a single attack family (`adam` -0.25, `apgd` -0.49), and crucially to a *single seed* (seed 42, $r = -0.24$, $n = 89$), so it is a cross-method structural confound rather than a restatement of SCRUB's cross-seed instability. The correlation is not a chance artifact: a 20,000-draw permutation null gives $p = 0.013$, it is sign-stable under leave-one-method-out (every single-family deletion leaves $r \in [-0.28, -0.20]$), it strengthens under outlier trimming ($r = -0.47$ on the central 90% of fragility), and—most importantly—it *replicates independently in all three seeds* (seed 42 $r = -0.24$, seed 123 $r = -0.35$, seed 456 $r = -0.63$), so it is not a one-off of the seed-42-heavy pool (`scripts/analysis/42_robustness_checks.py`). The consequence for the instrument is concrete: a globally brittle checkpoint inflates the apparent forget-vs-retain gap, so cross-checkpoint selectivity rankings are partly fragility rankings unless the absolute scale F is matched or reported alongside. The retain-only oracle null subtracts the oracle's *own* forget/retain asymmetry but does not equalize F between the unlearned model and the oracle, so it controls this confound only partially. We therefore read recovery-radius selectivity as a fragility-conditioned quantity: report F with every gap, and prefer fragility-matched comparisons (numbers in `results/analysis/metrics/fragility_confound.json`, script `scripts/analysis/40_fragility_confound.py`).

Why the confound exists: forget radius is super-elastic to fragility. The confound has a precise mechanism that the paper's own Theorem 1 makes falsifiable. If recovery radius were exactly $r \approx -m/L$ with a single global Lipschitz constant L, then L would cancel in the ratio r_R/r_F and selectivity would be fragility-*independent*: the naive theory predicts no confound. Fitting $\log r_F = a + b \log r_R$ across the 132 rows gives slope $b = 1.31$ with a suite-clustered bootstrap 95% CI of $[1.11, 2.04]$, excluding the no-confound value $b = 1$. The forget-class radius is thus *super-linearly* elastic to the global scale: as a checkpoint becomes globally more fragile (retain radius halves), the forget radius falls by $\approx 2^{1.31} \approx 2.5\times$, collapsing faster than retain controls and mechanically inflating apparent selectivity. The estimate is conservative—classic errors-in-variables attenuation biases the slope *toward* the no-confound value $b = 1$, so noise cannot manufacture $b > 1$ from a true $b = 1$—and it is stable under stress: a Theil–Sen robust slope gives $b = 1.74$, the central-90% fit gives $b = 2.42$, every leave-one-method-out fit stays in $[1.20, 1.58]$, and all three seeds reproduce $b > 1$ (seed 42 1.28, seed 123 1.64, seed 456 1.55). This locates exactly where the single-L reading of Theorem 1 breaks: forget and retain inputs sit in regions of systematically different local Lipschitz behavior, and that difference scales with global brittleness. The per-sample law above supplies the micro-mechanism: a checkpoint's selectivity tracks the *heterogeneity* of its forget margins. Where unlearning leaves a heterogeneous forget-margin distribution—some samples barely suppressed, sitting near the boundary—those samples recover at small radius and the row reads as selective; where it crushes every forget logit uniformly (e.g. SCRUB seed 123, all $\log p_t < -37$), there is no near-boundary tail and no selective gap. Across checkpoints, the within-checkpoint spread of $\log p_t$ correlates positively with selectivity ($r = +0.27$, $n = 55$), a modest but direction-consistent signal that ties the seed-level SCRUB instability to a measurable property of the forget-margin distribution rather than to residual class knowledge. The observational slope $b = 1.31$ pools methods of differing fragility, so we confirm it with a controlled within-method experiment (Appendix 3.9): holding SalUn fixed and varying only the base input-space smoothness through adversarial pre-training strength—a clean instrument that moves the global scale r_R over a $22\times$ range without a direct forget-suppression channel—yields a causal slope $\hat{b}_{\text{causal}} = 1.36$ (cluster-bootstrap 95% CI $[1.31, 1.84]$, ten checkpoints over two seeds), excluding the single-scale null $b = 1$ and matching the observational estimate (Figure 1D). Adapter-side weight decay is by contrast a first-stage null—the radius is governed by the base geometry, not the low-rank adapter (η^2 of $\log r_R$ is 0.25 by method but 0.002 by base seed and ≈ 0 by regularization)—which is exactly what identifies base smoothness as the correct knob. The super-elasticity is therefore a causal, not merely pooled, property of how the forget radius responds to fragility.

Cross-architecture replication. Both halves of this mechanism—the super-elastic slope $b > 1$ and the fragility confound $r < 0$—are not specific to the CIFAR-10/ViT-Tiny benchmark. We retrain six LoRA unlearning methods (SalUn, SCRUB, CE-ascent, Uniform-KL, feature-scrub, smoothed-margin) on two *new* architecture/dataset pairs—ResNet-18 (a convolutional network) and ViT-Small, both on CIFAR-100—and audit each with the identical recovery-radius protocol. Pooling the per-method points ($n = 11$; one SalUn/ViT-Small point is dropped because its forget radius collapses to exactly 0—maximally fragile and undefined on a log axis, which is itself direction-consistent) and refitting reproduces both effects: the cross-method elasticity is $\hat{b} = 1.98$ (case-resample 95% CI $[1.34, 3.13]$, with $b > 1$ in *every* leave-one-out fit, range $[1.73, 2.26]$), and the selectivity-versus-fragility confound is $r = -0.74$ (CI $[-0.92, -0.47]$, $r < 0$ in every leave-one-out fit, range $[-0.80, -0.68]$). This is an observational cross-method fit at a single seed per cell, so the defensible claim is the *direction*—super-elasticity ($b > 1$) and a negative fragility confound both hold across a CNN and a second ViT on a new dataset—not the precise exponent, which here (≈ 2) runs larger than the CIFAR-10 benchmark slope (1.31) and the controlled causal estimate (1.36). Figure 6 shows both panels; numbers in `results/analysis/metrics/crossarch_pooled.json` (script `scripts/analysis/51_crossarch_pooled.py`).

Which methods leak after fragility-matching. Removing the fragility trend ($S = a + c \log F + e$) leaves a fragility-adjusted selectivity e that ranks methods by leakage *beyond* what their brittleness explains (Table 6). Two features validate the adjustment. First, the retain-only *oracle null*—which has no forget-class knowledge by construction—falls to the bottom ($e = -0.66$, CI excluding 0), exactly where a correct adjustment must place it. Second, *SCRUB* sits at or below the oracle ($e = -0.76$): once fragility is controlled, its apparent selectivity vanishes entirely, mechanistically confirming the seed-level finding that SCRUB's leak is a brittleness artifact, not residual knowledge. The genuine leakers that survive the adjustment with CIs excluding zero are SalUn (+0.50), RURK (+0.44), and the smoothed-margin model (+0.46, small n=4)—the same rows the oracle-null audit flags as separating from the null—while BalDRO, CE-ascent, and Uniform-KL are indistinguishable from fragility-driven selectivity. Numbers and per-family CIs are in `results/analysis/metrics/elasticity_taxonomy.json` (script `scripts/analysis/41_elasticity_and_taxonomy.py`); families with $n < 4$ (e.g. OrthoReg) are excluded as underpowered.

8.3 Theorem 3: covering implies the mixed-surface failure

The patch, border-frame, and multi-patch results reported above are empirical context for Theorem 3. Patch-aware stage-2 closes the single-patch channel on ORBIT and CE-U, but the same defense becomes easier to recover under frame and multi-patch delivery, and mixed-surface stage-2—explicitly trained on multiple surfaces—does not reliably fix the transfer gap. Every coordinate is covered by some 32×32 patch (Lemma 1), so by Theorem 3 exact worst-case robustness across the family forces constancy at *any* budget—including the $\epsilon \approx 8/255$ the audits use, not only in a large-budget limit (Corollary 5). The empirical mixed-surface failure is consistent with this obstruction, but the theorem does not rule out average-case or data-manifold robustness under realistic sampled mask distributions; that remains the appropriate local-delivery defense target.

8.4 Theorem 4: predicted vs. measured certified radius

The smoothed objective of Eq. (13) is implemented as a hinge on the Monte-Carlo smoothed target margin and trained end-to-end. On seed 42, starting from a base ViT-Tiny at 94.2% validation

Method family	n	raw selectivity	fragility-adjusted e (CI95)
SalUn	20	1.39	$+0.50$ $[+0.08, +0.91]$
Smoothed-margin	4	0.91	$+0.46$ $[+0.18, +0.73]$
RURK	12	1.26	$+0.44$ $[+0.21, +0.62]$
BalDRO	16	0.83	$+0.05$ $[-0.12, +0.25]$
CE-ascent	17	0.79	-0.02 $[-0.06, +0.01]$
ORBIT	30	0.61	-0.14 $[-0.21, -0.05]$
Uniform-KL	6	0.77	-0.19 $[-0.57, +0.20]$
Oracle null	3	-0.01	-0.66 $[-0.81, -0.51]$
SCRUB	17	-0.03	-0.76 $[-0.98, -0.59]$

Table 6: Fragility-matched selectivity taxonomy. Raw selectivity $S = \log(\tilde{r}_R/\tilde{r}_F)$ is confounded by global fragility; the adjusted e is the residual after regressing S on $\log F$. The oracle null and SCRUB fall to the bottom—validating the adjustment and confirming SCRUB's leak as a brittleness artifact—while SalUn, RURK, and smoothed-margin survive as genuine leakers (CIs excluding 0). Bootstrap 95% CIs over rows; small-n families (< 4) omitted.

accuracy, a 10-epoch smoothed-margin LoRA fine-tune ($\sigma = 0.10$, $n_{\text{noise}} = 4$, $\gamma = 1.0$) drives forget accuracy 95.7% $\to$ 12.0% while retain accuracy is preserved (93.9% $\to$ 93.8%). Seeds 123 and 456 reach forget accuracy 12.1% and 9.3% with retain preserved. The objective is designed to reduce the noisy forgotten-class event, but the certificate is established only post hoc by the Monte-Carlo upper confidence bound in Theorem 4. The exact theorem applies to the binary forgotten-class detector under *unclipped* Gaussian noise; noisy-input clipping during training or auditing is an implementation choice and must be interpreted as a clipped-distribution diagnostic rather than the exact Cohen–Rosenfeld–Kolter certificate. Under the native L_2 audit, the reported detector-radius scale is $R = 0.193$, while measured deterministic-model L_2 recovery radii remain above it across seeds. This is evidence that the deterministic recovery radius is compatible with the detector certificate, not a direct proof of the smoothed detector's recovery radius. The L_∞ comparison is moved to Appendix F: it is a cross-norm stress diagnostic comparing an L_2 detector bound, after $\sqrt{d}$ conversion, to a base-model L_∞ recovery audit. Because $d = 150{,}528$, the converted scale is only ≈ 0.0005, so that plot is a caveat about norm mismatch rather than validation of Theorem 4.

Native L_2 recovery audit. The $\sqrt{d}$ deflation above is an artifact of comparing an L_2 detector-radius scale against an L_∞ audit. Re-running the deterministic recovery audit under an L_2 budget gives a forget median recovery radius of 0.46–0.57 across seeds 42/123/456, above the reported L_2 detector-radius scale $R = 0.193$. The L_2 audit also sharpens the selectivity picture: at the L_2 budget the forget and retain *success rates* no longer saturate, separating to 0.84 (forget) versus 0.03 (retain)—a 0.82 rate gap. This native-norm diagnostic is the cleanest empirical companion to the detector certificate; the converted L_∞ plot is only a caveat about metric mismatch.

Direct smoothed-detector certificate. Table 7 reports the Theorem 4 certificate *directly* on the binary smoothed forget-detector g_t, rather than inferring it from the deterministic audit. Every noise draw predicts a non-forget class ($\hat{p}_{\neg t} = 1$ at $m = 256$), so the Clopper–Pearson lower bound on the non-forget probability is $\underline{p}_{\neg t} = 0.973$ and the certified radius $R = \sigma\,\Phi^{-1}(\underline{p}_{\neg t}) = 0.193$ saturates identically across all 64 forget samples and all three seeds; equivalently the upper bound on the forgotten-class-top-logit event is $\tau = 1 - \underline{p}_{\neg t} \approx 0.027$. The deterministic native-L_2 forget recovery medians sit a factor $\sim$2.4–3.0 above R (no L_2 violation) and are the empirical companion to the

seed	σ	m	n	τ (CP)	cert. frac.	detector R	det. L_2 med.
42	0.10	256	64	0.027	1.00	0.193	0.488
123	0.10	256	64	0.027	1.00	0.193	0.461
456	0.10	256	64	0.027	1.00	0.193	0.570

Table 7: Smoothed forget-detector certificate (Theorem 4), $\sigma = 0.10$, $\alpha = 10^{-3}$, *unclipped* Gaussian noise (the exact Cohen–Rosenfeld–Kolter setting). Columns: noise draws m; audited forget samples n; threshold τ, the Clopper–Pearson upper bound on the forgotten-class-top-logit event q_t; fraction of forget inputs with a positive certificate; the L_2 detector radius $R = \sigma\,\Phi^{-1}(1-\tau)$; and the deterministic native-L_2 forget recovery median. The certificate is identical across seeds because every noise draw is non-forget, while the deterministic median varies; retain controls give an identical certificate. Numbers from `smoothed_radius_seed_*_sigma_0p1_n256_*.json`.

σ	forget acc	retain acc	L_2 detector radius R
0.05	0.2%	93.7%	0.097
0.10	12.0%	93.8%	0.193
0.25	66.4%	93.9%	0.483
0.50	83.5%	94.1%	0.967

Table 8: Certification–utility frontier for the smoothed-margin objective (seed 42, $n_{\text{noise}} = 256$, $\alpha = 10^{-3}$). Larger σ buys a larger L_2 detector-radius scale but degrades clean forgetting; retain accuracy is preserved across the sweep.

certificate, not a substitute for it. Holding this model fixed and raising the *certification* noise—a distinct role from the *training* scale swept in Table 8—to $\sigma = 5.0$ leaves the forgotten class off the top logit in *every* one of 65,536 noisy forwards ($\hat{p}_{\neg t} = 1.0$ over 256 samples $\times$ 256 draws, unclipped, seed 42), so the radius scales linearly to $R = 9.66$ with the certified fraction pinned at 1.0. The saturated certificate columns therefore reflect global suppression of the forget-class prior across input space, not a reporting artifact; the seed-to-seed signal is carried by the deterministic median.

Certification–utility frontier. Theorem 4 exposes a tunable tradeoff through the smoothing scale σ: a larger σ inflates the detector-radius scale ($R = \sigma\,\Phi^{-1}(1-\tau)$, with τ the upper confidence bound on the noisy forgotten-class event), but the same noise that buys the certificate also washes out the clean forgetting signal. Table 8 sweeps σ on seed 42. The L_2 detector radius grows linearly with σ (from 0.10 at σ=0.05 to 0.97 at σ=0.50), but clean forget accuracy degrades sharply (from 0.2% to 83.5%) while retain accuracy stays preserved throughout. A wide radius therefore does *not* imply good unlearning: at σ=0.50 the smoothed classifier carries a large radius while the underlying model has effectively stopped forgetting. The useful operating regime is small σ (0.05–0.10), where forgetting is strong and the radius is modest—which is why the headline run uses σ=0.10.

Cross-architecture check. To check that the smoothed objective is not only a ViT-Tiny/CIFAR-10 artifact, the same objective is run on a ViT-Small CIFAR-100 base checkpoint (seed 42, forgotten class 0). The base starts with meaningful class-0 behavior (forget accuracy 78.0%, retain accuracy 54.7%). A 10-epoch smoothed-margin LoRA fine-tune drives forget accuracy to 0.0% while retaining 47.7% clean accuracy, so this row is a cross-dataset feasibility check with a nontrivial retain drop, not a utility-preserving result on the same footing as CIFAR-10. The native L_2 recovery audit puts measured recovery radii well above the reported detector-radius scale (forget median 1.39, retain

Method	Architecture	Forget acc (before→after)	Retain acc (before→after)	Robustness audit
Smoothed-margin	ViT-Small	78.0 → 0.0	54.7 → 47.7	certified L_2 $R = 0.193$
RURK	ViT-Small	88.1 → 0.0	94.3 → 93.7	AWP recovery gap 0.9%
SalUn	ResNet-18	92.2 → 0.0	94.6 → 91.5	AWP recovery gap 0.0%

Table 9: Single-seed CIFAR-100 cross-check (forgotten class 0). All three methods drive forget accuracy to 0%. RURK and SalUn largely preserve retain accuracy and resist a weight-perturbation (AWP) recovery audit at $\epsilon = 10^{-3}$; the smoothed-margin row uses a lightly-trained base and has a larger retain drop, but is the only row carrying the smoothed detector-radius audit. Numbers in `cross_arch_cifar100_summary.json`.

median 1.41), but the forget and retain recovery radii are nearly identical under both L_2 and L_∞ audits; the row therefore supports objective/certificate feasibility, not selective leakage. To check that the pipeline is not tied to one architecture, two literature methods are also run on CIFAR-100 from better-trained bases: RURK on ViT-Small and SalUn on a *ResNet-18*. These rows drive forget accuracy to 0.0% while largely preserving retain accuracy (RURK 94.3%→93.7%; SalUn 94.6%→91.5%), and both resist an adversarial weight-perturbation (AWP) audit—which perturbs weights inside an $\epsilon = 10^{-3}$ ball and re-measures forget accuracy—keeping forget accuracy at or below 0.9% under the perturbation. Table 9 collects the three single-seed rows; they broaden the benchmark beyond CIFAR-10/ViT-Tiny but do not establish a multi-seed cross-dataset claim.

9 Defense Discussion

9.1 Feature-subspace stage-2

The strongest post-hoc defense candidate in the current benchmark is a feature-subspace stage-2 finetune. Rather than modifying the full unlearning objective from epoch 0, it starts from an already-unlearned adapter and suppresses attacked forget-sample features inside a learned recovery subspace. This avoids the instability of earlier logit-margin and teacher-distillation defenses on BalDRO, but the evidence remains point-estimate evidence rather than a fully paired confidence-interval result.

9.2 Why the gains are small

The absolute changes in reported radii are numerically small because the benchmark itself is small: CIFAR-10 under an L_∞ budget on 224×224 inputs yields radii on the order of a few thousandths. The right question is therefore relative: does the defense move the forget-vs-retain frontier without collapsing clean utility? The point estimates move in the right direction on BalDRO, SalUn, and ORBIT, but the BalDRO shift is too small to emphasize without uncertainty intervals; the defensible reading is a modest ORBIT/SalUn signal plus a tiny BalDRO sanity check.

9.3 What does not generalize

The defense is not universal. RURK remains a hard case under both attack and defense. SCRUB behaves like generic adversarial fragility rather than a clean forgetting-leak regime, so stronger attacks on SCRUB do not necessarily imply a meaningful unlearning failure mode. The newer delivery-surface results make this sharper: patch-aware defenses often fail to transfer to border-frame or multi-patch delivery, and the mixed-surface stage-2 defense does not reliably fix that.

10 Proposed Defense Objective

The experiments suggest that clean-point forgetting is too weak a training target. A more principled direction is targeted-margin adversarial unlearning:

$$\min_{\theta} \mathbb{E}_{x\in R} L_{\text{retain}}(x;\theta) + \lambda_f \mathbb{E}_{x\in F} L_{\text{forget}}(x;\theta) + \lambda_m \mathbb{E}_{x\in F} \max(0,\ m_t(x+\delta^*(x)) + \gamma)\,, \tag{15}$$

where F is the forget set, R is the retain set, m_t is the forgotten-class target margin, $\delta^*(x)$ is an inner-loop or amortized attack, and γ is a safety margin. This objective is purely empirical: it minimizes a worst-case inner attack and carries no certificate. Theorem 4 replaces the inner attacker $\delta^*(x)$ with Gaussian smoothing of the target margin, Eq. (13), yielding what appears to be an early unlearning objective with a per-input input-space randomized-smoothing lower bound on forgotten-class recovery radius. The end-to-end result in Section 8 (seed 42: forget 95.7% $\to$ 12.0%, retain preserved, L_2 detector-radius scale 0.193; replicated across seeds 42, 123, and 456) is the realized version of this objective. The implemented feature-subspace stage-2 defense can be viewed as a computationally cheaper, uncertified approximation to the same idea: defend against locally recoverable targeted behavior, not just against clean forget accuracy.

11 Discussion

This work began as a multi-regime red-team study, and Section 8 turns that benchmark into empirical checks for the theory of Section 3: the Lipschitz–radius scatter supports Theorem 1 as a ranking diagnostic, the Pinsker/oracle-null audit checks Theorem 2's selective-leakage interpretation, the patch/frame/multi-patch transfer results are consistent with Theorem 3, and the smoothed-margin run realizes Theorem 4. Read together, the evidence supports a non-universal empirical picture. Strong per-example conditional recovery is the main general threat; universal perturbations are mostly weak on the stronger checkpoints; conditional patches expose an additional local attack surface on some methods; border-frame and multi-patch delivery show that exact worst-case local-delivery transfer is the wrong target; robust-base methods define a utility-vs-recovery frontier rather than a binary robust point; and feature-subspace stage-2 provides modest point-estimate gains on several targets. Later defense variants remain method-specific: teacher-guided stochastic CVaR helps ORBIT slightly but not CE-U, while patch-aware stage-2 gives the first clearly positive CE-U result in the patch regime (68.8%$\to$43.8% at 32×32, seed 42) with no retain-side failure, though it does not help on the full-image conditional attack. Uniform KL reinforces this framing: seed 42 is vulnerable under both the older prompt/probe family and the newer conditional audit, while seed 123 stays flat under the same fast conditional attack.

12 Limitations

The paper does not claim a universally stronger new optimizer than strong first-order attacks such as `adam_margin` or `apgd_margin`. The amortized short-budget attack is partial and target-dependent. The defense gains are modest and currently demonstrated on a benchmark that is intentionally controlled rather than broad: mostly CIFAR-10 ViT checkpoints, often with LoRA-based unlearning implementations, plus a smaller number of full-finetune and cross-architecture checks. This makes the benchmark useful for discovering failure modes and transfer gaps, but not yet a direct proxy for most industry deletion settings. Finally, some regimes are better understood as negatives or controls than as headline wins.

A reproducibility caveat is stated up front rather than left to the appendix. The distributed SalUn baseline checkpoint does *not* reproduce its reported clean-forget rate (it predicts the forgotten class on 28–53% of clean forget inputs from the published `checkpoints` branch); the SalUn training recipe does reproduce, and the regenerated 3-seed baseline supersedes the shipped checkpoint. The plain BalDRO, ORBIT, and RURK baseline recipes were never committed (only `_smoke` variants exist), so BalDRO is re-reported as a recipe-scaled *reconstruction* and the ORBIT and RURK baselines—including ORBIT, a flagship diagnostic row—remain single-run artifacts not yet independently reproduced. Appendix I and the reproducibility appendix give the full audit; the takeaway is that the tracked result JSONs and recipes, not the distributed checkpoints, are the primary reproducibility surface.

The theory carries its own caveats. (i) Corollary 2 is rigorous for the bounded probability margin (constant 2, no logit bound), but the Lipschitz constant L it needs is estimated empirically by a logit-margin Jacobian proxy $\hat{L}$ at audit points; because this proxy is not a certified global upper bound, the scatter gives a heuristic lower-bound estimate and ranking rather than a valid certificate floor. The pointwise inequality itself, however, is validated directly at the sample level (median $\mathrm{corr}(\hat{r}, \log p_t) = -0.64$ over 58 checkpoints; Section 8), so the caveat is about the unmeasured global L, not about Eq. (5). (ii) Pinsker's inequality in Theorem 2 is loose; the tighter Bretagnolle–Huber bound is reported alongside it, but neither is a finite-sample hypothesis test at the small per-row audit sizes ($n_F = n_R$ from 4 to 32). A larger-n follow-up ($n = 128$) with bootstrap confidence intervals and a retain-only oracle null (Section 8) backs the main selective rows and shows that the forget-vs-retain gap is not a baseline artifact; it also reveals that the *sign* of SCRUB's selectivity is seed-unstable (selective on 42, anti-selective on 123, parity on 456, with two of three CIs excluding parity in opposite directions), so neither a selective-leak nor a generic-fragility label is robust for SCRUB. More broadly, a benchmark-wide audit (132 rows, 57 checkpoints; Section 8) shows the selectivity metric carries a modest but robust *fragility confound*—signed selectivity correlates with the oracle-free retain-control radius scale at $r = -0.25$ (suite-clustered CI $[-0.44, -0.09]$, holding within a single seed)—so selectivity should be read as fragility-conditioned and cross-checkpoint comparisons should be fragility-matched; the oracle null controls this only partially. (iii) Theorem 3 holds at *any* budget, including the audit budget, but it constrains only *worst-case* invariance; average-case robustness over sampled masks remains open and is the relevant distributional defense target. (iv) Theorem 4 is an L_2 result for a binary smoothed detector. Converting that scale to L_∞ at $d = 150{,}528$ incurs the $\sqrt{d} \approx 388$ penalty and yields a numerically tiny converted bound (≈ 0.0005), so the native L_2 audit is the main diagnostic companion and the L_∞ plot is only a caveat. The main end-to-end smoothed-margin result replicates across seeds 42, 123, and 456 with a consistent 2.7–4.3× selective gap on CIFAR-10/ViT-Tiny; single-seed cross-architecture CIFAR-100 rows (smoothed-margin and RURK on ViT-Small, SalUn on ResNet-18; Table 9) broaden the benchmark but are not a multi-seed cross-dataset claim. The structural findings travel further than the feasibility rows: the super-elasticity ($b > 1$) and the fragility confound ($r < 0$) both replicate directionally on ResNet-18 and ViT-Small / CIFAR-100 ($n = 11$ pooled, every leave-one-out fit; Section 3), but this remains an observational cross-method fit at a single seed per cell, so it pins the sign, not the exponent. Each caveat is a concrete next step, not a hidden assumption.

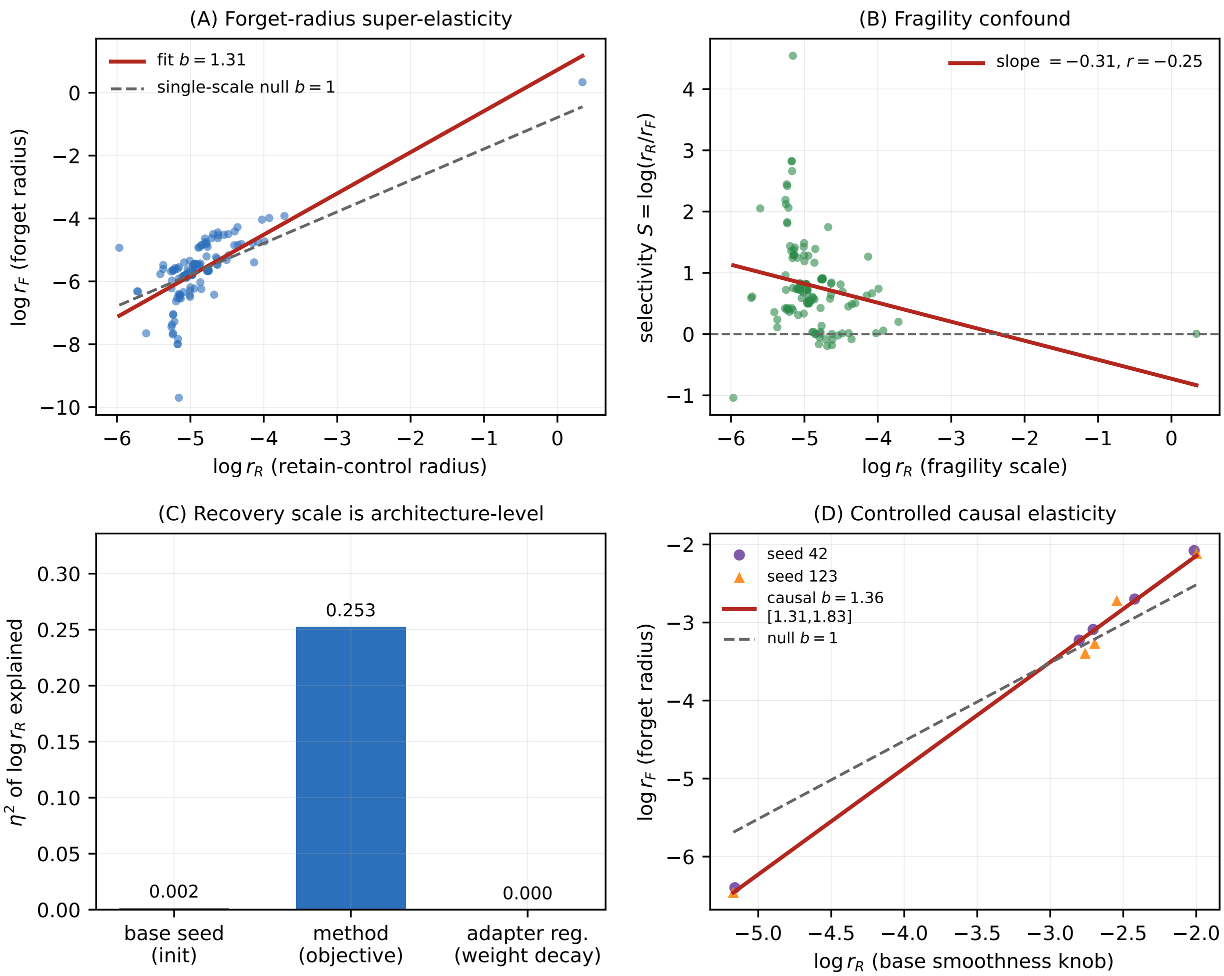

Figure 1: **The fragility confound and where the recovery scale lives** (132 audited checkpoints, one architecture, three base seeds, twelve method families). **(A)** Forget-radius super-elasticity: a log–log fit of the forget radius on the retain-control radius has slope $\hat{b} = 1.31$, exceeding the single-scale null $b = 1$ of Theorem 1 (Proposition 1). **(B)** Signed selectivity $S = \log(r_R/r_F)$ falls with the global fragility scale $\log r_R$ ($r = -0.25$): globally brittle checkpoints look spuriously selective. **(C)** Variance of $\log r_R$ explained (η^2): the base random seed explains essentially none (0.002) and adapter-side weight decay none (the first-stage null of Section 3.9), while the unlearning objective explains 0.25—the global recovery scale is an architecture-level property, not an initialization or regularization artifact. **(D)** Controlled *causal* elasticity: holding the unlearning method fixed and varying only the base input-space smoothness (adversarial pre-training strength) moves r_R over a $22\times$ range and gives within-method slope $\hat{b} = 1.36$ (cluster-bootstrap 95% CI $[1.31, 1.84]$, two seeds, ten checkpoints), excluding the single-scale null $b = 1$ and matching the observational $b = 1.31$ of panel (A).

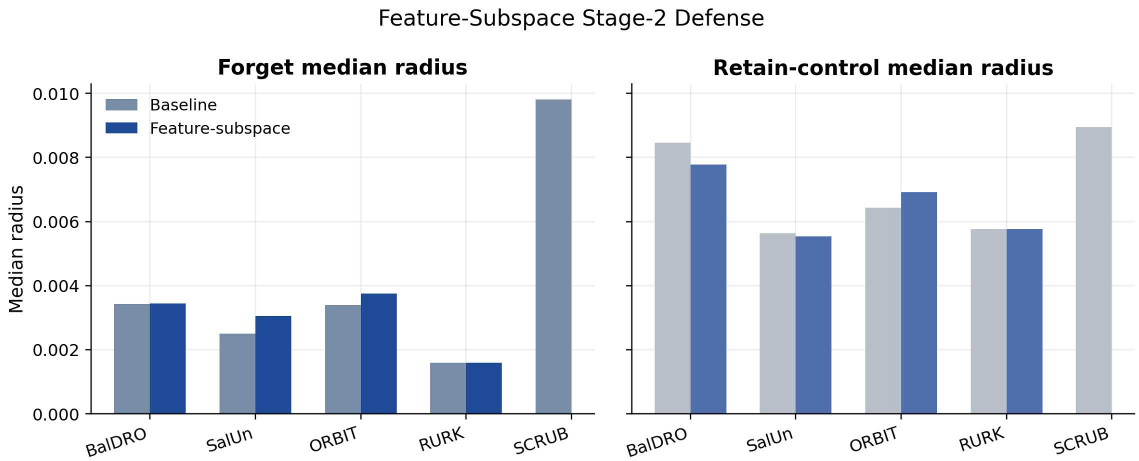

Figure 2: Feature-subspace stage-2 tradeoff. Larger forget medians indicate stronger resistance to conditional recovery. The plotted point estimates improve most clearly on ORBIT and SalUn, shift only slightly on BalDRO, and do not move RURK. Controlled paired confidence intervals require re-auditing matched-attack baselines from reproducible checkpoints and remain future work.

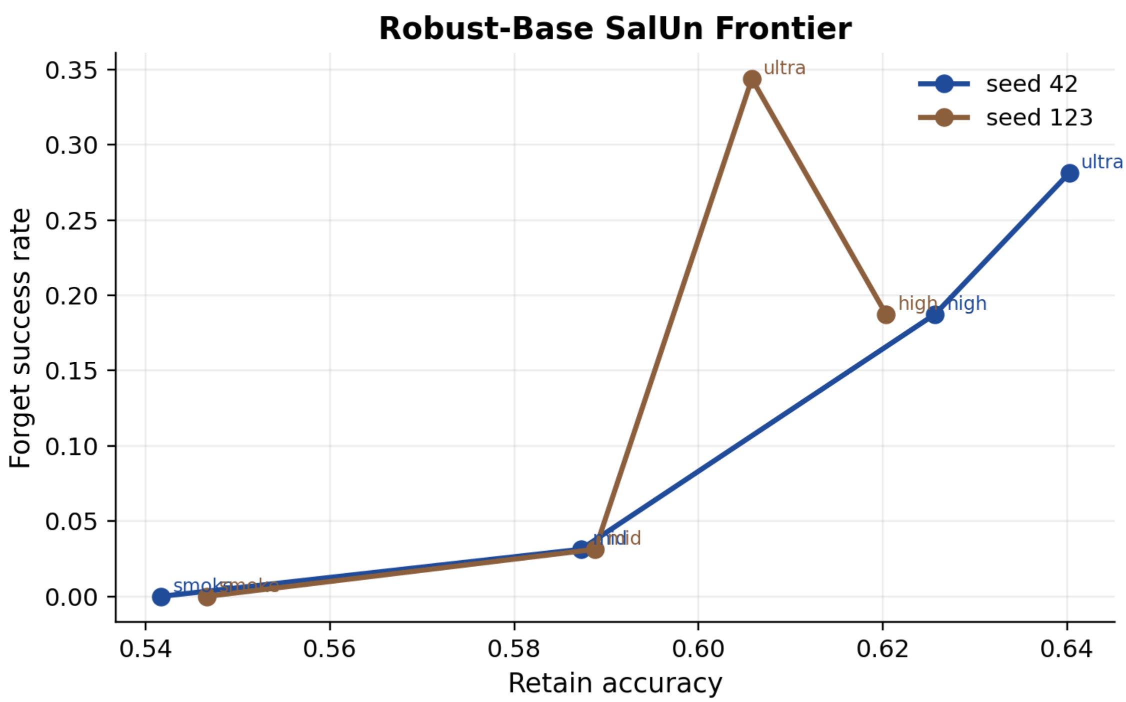

Figure 3: Robust-base SalUn frontier. Lower-utility robust-base checkpoints suppress conditional recovery more strongly, but leakage returns as retain utility improves.

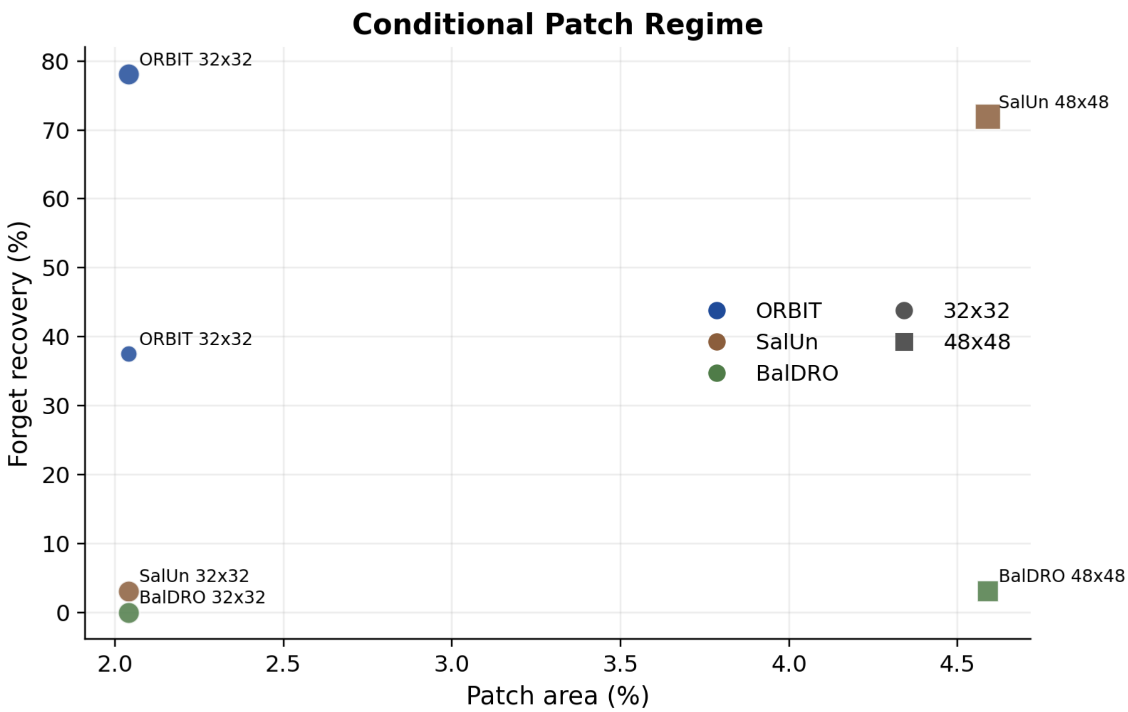

Figure 4: Conditional patch regime. ORBIT admits strong low-area patch recovery, SalUn requires a larger patch and incurs more retain damage, and BalDRO is effectively negative under the same patch family.

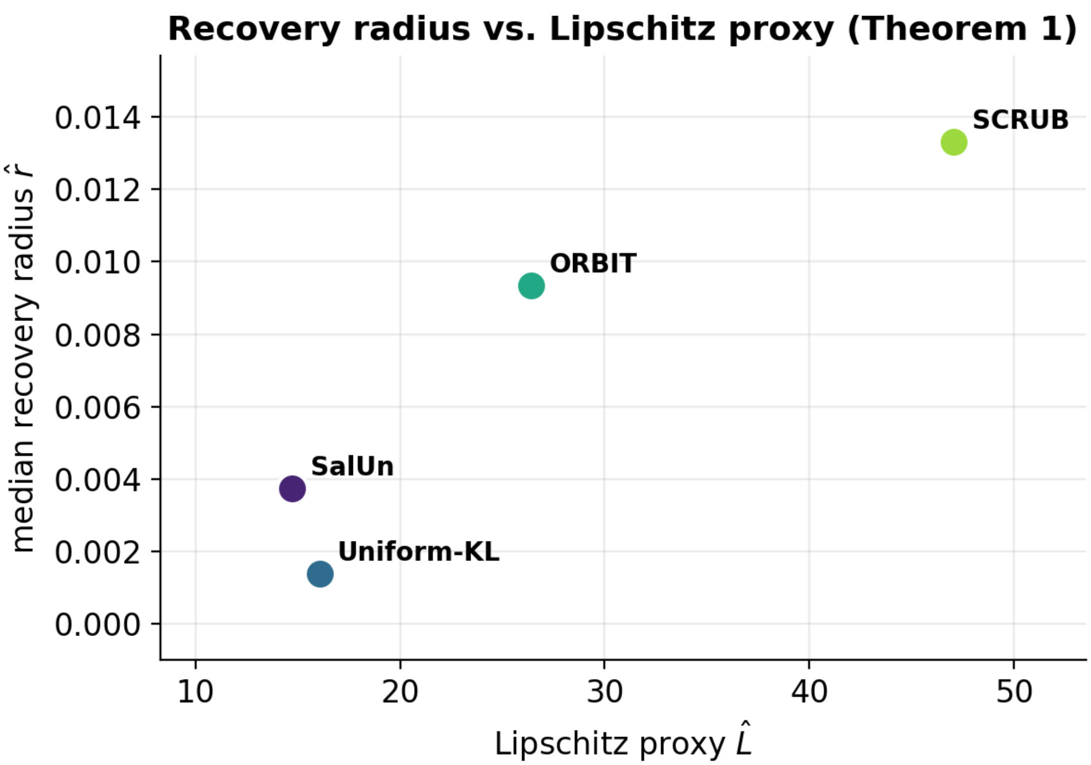

Figure 5: Theorem 1 ranking diagnostic. Per-method aggregate Lipschitz proxy $\hat{L}$ versus aggregate median recovery radius $\hat{r}$, summarized from the 40 joined audit rows. The plot orders methods by $\hat{L}$ rather than drawing the pointwise bound $\hat{r} \geq -\bar{m}/\hat{L}$, which would need a per-method oracle margin $\bar{m}$ unavailable to an oracle-free auditor; it visualizes the Corollary 2 ranking, not a tightness test.

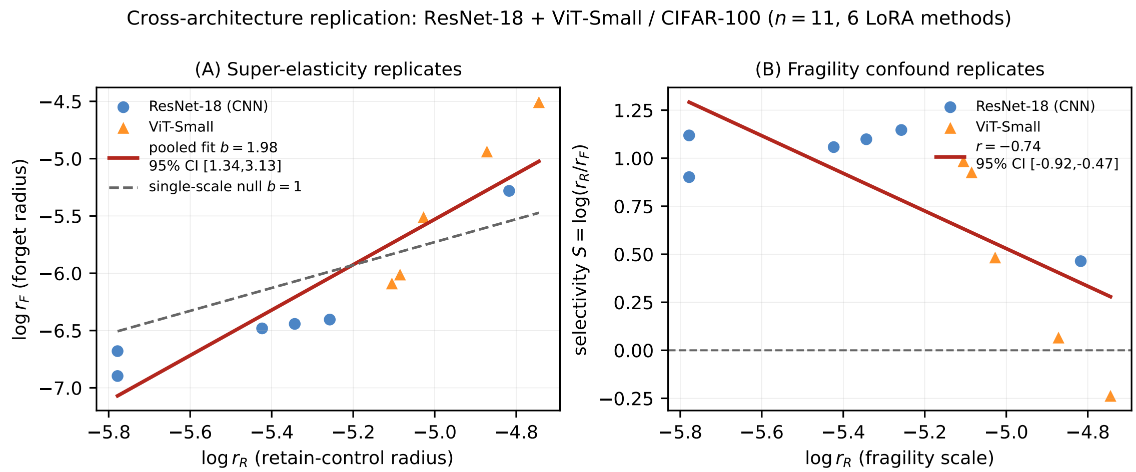

Figure 6: Cross-architecture replication on two new architecture/dataset pairs (ResNet-18 and ViT-Small, both CIFAR-100; six LoRA unlearning methods each, $n = 11$ pooled). **(A)** Forget-radius super-elasticity: the pooled log–log fit has slope $\hat{b} = 1.98$ (95% CI $[1.34, 3.13]$), steeper than the single-scale null $b = 1$ (dashed), with $b > 1$ in every leave-one-out fit. **(B)** The selectivity-versus-fragility confound reappears with $r = -0.74$ (95% CI $[-0.92, -0.47]$), $r < 0$ in every leave-one-out fit. Both effects hold across a CNN and a second ViT, fixing the direction of the mechanism rather than the exact exponent (single seed per cell).

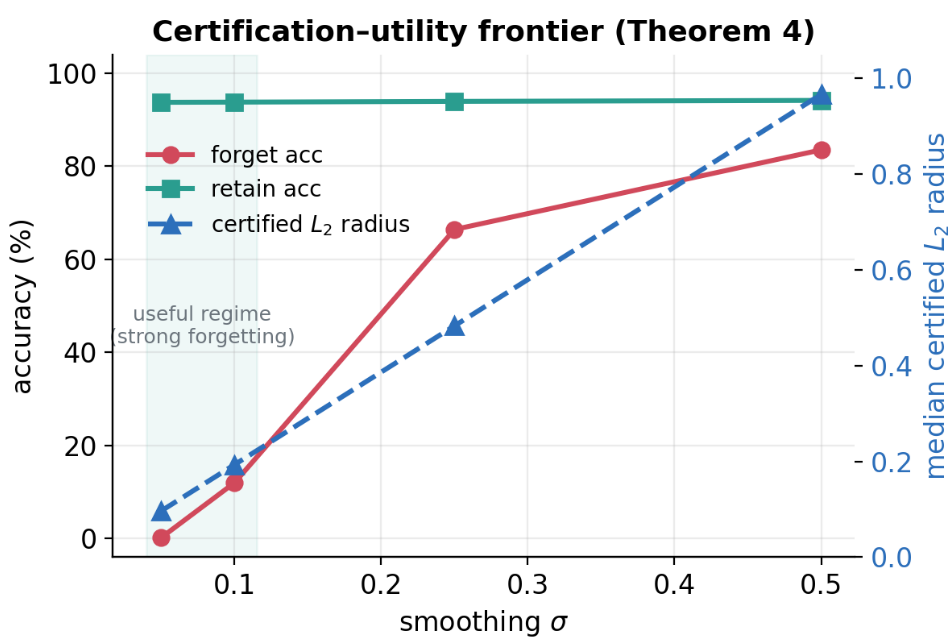

Figure 7: Certification–utility frontier (Theorem 4, seed 42). The L_2 detector-radius scale (right axis) rises linearly with σ, while clean forget accuracy (left axis) collapses upward as the smoothing noise overwhelms the margin objective; retain accuracy is flat. Strong forgetting and a large radius are in tension.

13 Conclusion

Machine unlearning cannot be evaluated reliably with clean forget accuracy alone. This paper makes the per-example recovery radius the central object and shows it supports an operational theory: a Lipschitz recovery–certification gap that gives a necessary condition for certificate claims (Theorem 1), a Pinsker bound plus oracle-null calibration for selective leakage (Theorem 2), refined by a benchmark-wide fragility confound—globally brittle checkpoints inflate apparent selectivity, so a fragility-matched taxonomy is needed to tell genuine leakers (SalUn, RURK) from brittleness artifacts (SCRUB), a covering-family impossibility for exact worst-case local-delivery invariance (Theorem 3), and a smoothed objective with a per-input L_2 recovery-radius lower bound for a binary forget detector (Theorem 4). Across the benchmark, the dominant leakage channel is strong per-example conditional recovery, but local patch, frame, and multi-patch attacks expose additional method-specific surfaces that do not reduce to a single universal vulnerability. The defense picture is similarly non-universal: feature-subspace stage-2 gives modest point-estimate gains on several rows, patch-aware training helps ORBIT and CE-U locally, and average-case delivery-surface transfer remains open. The benchmark is the experimental support for this theory, not a substitute for broader multi-dataset evaluation.

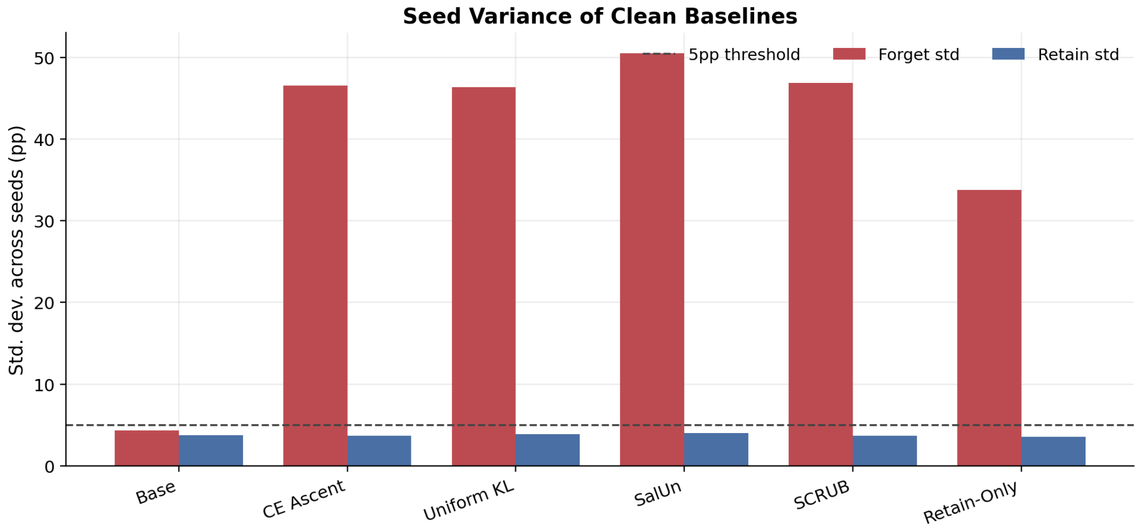

Figure 8: Seed variance of clean baselines. Forget-side variance is large for several methods even when retain-side variance remains small.

A Appendix: Seed Variance and Baseline Stability

The clean baseline regime itself is highly variable across seeds for several unlearning methods, especially on forget accuracy. Figure 8 summarizes this instability. This is one reason the paper emphasizes attack-conditioned evaluation rather than clean forget accuracy alone.

B Appendix: Robust-Base Frontier

Figure 3 (main text) summarizes the robust-base SalUn operating-point story. At low utility, recovery can be suppressed strongly or disappear on the sampled audit. As retain accuracy increases, conditional recovery re-emerges. This makes robust-base training a frontier choice rather than a single robust point.

C Appendix: Conditional Patch Regime

The patch regime is method-dependent. ORBIT is the flagship positive case with a 32×32 conditional patch covering 2.04% of the image, strong forget recovery, 0% retain-to-forget success, and limited retain accuracy drop. SalUn is patch-vulnerable only at a larger 48×48 patch budget. BalDRO is effectively negative at the tested budgets. Figure 4 (main text) summarizes the patch success rates across methods and seeds.

D Appendix: Delivery-Surface Transfer

The later delivery-surface checks separate one-patch wins from broader local-delivery robustness. Under the conditional frame attack, the ORBIT patch-stage2 row and CE-U combined stage-2 both become easier to recover, and CE-U also incurs retain-side failure. Under conditional multi-patch delivery, the ORBIT patch-stage2 row transfers only weakly, while CE-U combined stage-2 and faithful full-ft SalUn patch-stage2 become easier to recover. Mixed-surface stage-2 was designed to fix this transfer problem, but in practice it only gives narrow gains and often worsens frame or multi-patch robustness. Delivery-surface transfer therefore remains an open average-case defense problem rather than solved by the current stage-2 families.

E Appendix: Negative and Hard Cases

RURK remains a hard case for both the amortized attack family and the feature-subspace defense. SCRUB is useful as a negative control of a sharper kind: its forget-vs-retain selectivity is not merely weak but *sign-unstable* across seeds. Under the matched n=128 `adam_margin` oracle-null audit it is selective on seed 42 (2.04× easier, bootstrap CI $[1.85, 2.38]$), significantly *anti*-selective on seed 123 (forget *harder* than retain, 0.83×, CI $[0.74, 0.93]$), and at parity on seed 456 (1.01×, CI $[0.82, 1.21]$), all against a flat oracle null. Two of three seeds have CIs that exclude parity, in opposite directions, so SCRUB cannot be assigned a stable selective-leak or generic-fragility label; the seed-42 selective reading in particular is not a reliable leak. Several later defense variants also belong in this bucket: mixed-surface stage-2, frame-transfer failures, and the faithful full-ft SalUn mixed-surface row are all informative mainly because they show what does not transfer.

F Appendix: Cross-Norm Diagnostic

Figure 9 is retained only as a norm-mismatch diagnostic. The x-axis is the L_∞ scale obtained by converting an L_2 detector radius by the $\sqrt{d}$ factor, while the y-axis is a base-model L_∞ recovery audit. Points below the line should not be interpreted as failures of the detector certificate; they show why the native L_2 audit and Table 7, not this converted plot, are the appropriate companion evidence for Theorem 4.

G Appendix: Claim–Evidence Map

Table 10 summarizes the paper's strongest claims and the evidence class supporting each. The goal is to separate theorem statements, measured benchmark findings, diagnostics, and pilot observations.

H Appendix: Reproducibility Artifacts

The paper draws from repository revision `8de2f0ecdf8a` for the result JSONs and figure sources used in this revision. The main summaries and generated figures live in:

- `results/analysis/feature_subspace_tradeoff.md`
- `results/analysis/recovery_tradeoff_summary.md`
- `results/analysis/conditional_patch_summary.md`
- `results/analysis/paper_results_outline.md`
- `manifest.json`, regenerated by the repository manifest audit script (which also writes an ignored human-readable report)
- `results/analysis/figures/`

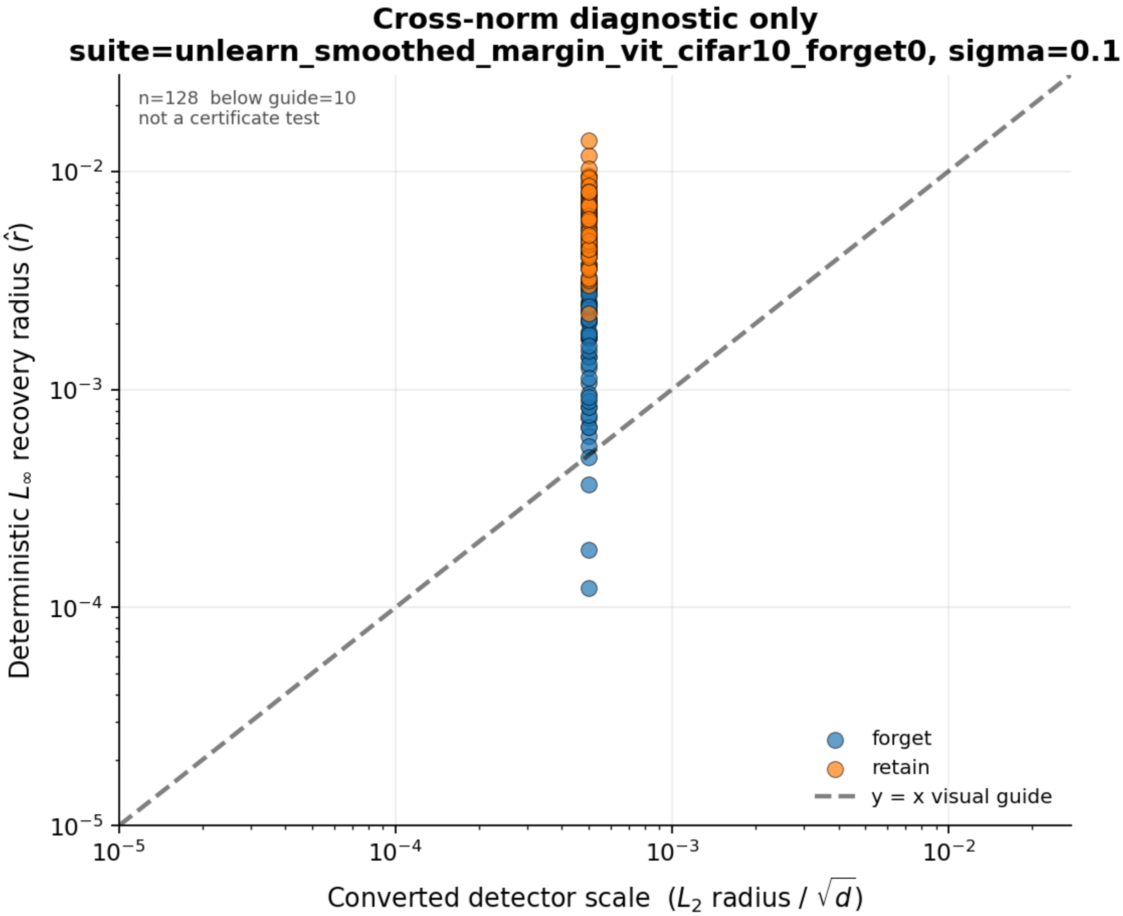

Figure 9: Cross-norm stress diagnostic for the smoothed-margin row (seed 42). This is not a certificate test: it compares a dimension-converted L_2 detector-radius scale to a deterministic base-model L_∞ recovery audit.

All figures and tables in this paper are generated from JSON result files committed alongside the code, together with the audit scripts needed to reproduce each attack regime. Full checkpoint reproduction is heavier: the trained checkpoints live on a separate `checkpoints` branch (base, oracle, and several unlearn adapters across seeds 42, 123, and 456), supplemented by theory-era and stage-2 checkpoints that are regenerated locally from the training scripts. Not all eight benchmark methods are checkpoint-covered across all three seeds on the main branch; the tracked JSON artifacts, not the checkpoints, are the primary reproducibility surface.

Checkpoint reproducibility and a regenerated SalUn baseline. A reproducibility audit of the distributed baseline checkpoints found that the shipped SalUn forget-class-0 LoRA adapter does not reproduce its reported clean-forget rate: audited from the published `checkpoints` branch it predicts the forgotten class on 28–53% of clean forget inputs (depending on the saved sub-adapter), not the near-zero rate of Table 3. The training *recipe* is intact, however: retraining SalUn from the committed configuration (`configs/suites/repro_salun.yaml`; objective `salun`, rank 8, 50 epochs, lr $= 10^{-3}$) reproduces proper unlearning across all three seeds—clean forget accuracy 0.0% with a forget recovery median of 0.00349, 0.00368, 0.00417 (L_∞, `adam_margin`, 40 steps) for seeds

Claim	Evidence class	Evidence boundary
Recovery radius is an oracle-free audit primitive.	Definition + audit protocol	The radius estimator does not require a retrained oracle; oracle-null calibration is optional and used only to interpret selectivity.
Theorem 1 gives a recovery–certification necessary condition.	Theorem + diagnostic	The theorem is exact for the stated Lipschitz constant; Figure 5 uses a local logit-margin proxy and is only a ranking diagnostic.
Large forget-vs-retain rate gaps imply a variational signal.	Theorem + measured rates	Pinsker lower-bounds KL only when a gap exists; it does not certify low leakage when the measured gap is near zero.
The oracle-null audit separates residual leakage from target-label asymmetry.	Measured control	Larger-n retain-only oracle audit shows near-zero baseline asymmetry; it is an operational null, not an information-theoretic zero-knowledge proof.
Worst-case delivery-surface invariance over a covering family is impossible for nonconstant classifiers.	Theorem	The result applies to exact global worst-case invariance with submask-supported perturbations; average-case sampled-mask robustness remains open.
The smoothed objective yields an L_2 detector certificate.	Post-training certificate	The certificate is for a binary smoothed forgotten-class detector under unclipped Gaussian noise; deterministic L_2 recovery is companion evidence, not the proof.
Feature-subspace stage-2 gives positive shifts on several rows.	Point-estimate benchmark	Clearest on ORBIT and SalUn; BalDRO is a tiny positive shift and needs paired confidence intervals before stronger statistical language.
CE-U shows selective full-image recovery.	Pilot	Seed-42, $n = 8$, two-step audit; suggestive only and not ranked against full-audit rows.
CIFAR-100 checks broaden scope.	Single-seed feasibility	Useful cross-dataset/architecture checks, not a multi-seed cross-dataset benchmark claim.
Super-elasticity ($b > 1$) and the fragility confound ($r < 0$) generalize across architectures.	Observational cross-method replication	Six LoRA methods on ResNet-18 and ViT-Small / CIFAR-100 ($n = 11$ pooled) reproduce $b > 1$ and $r < 0$ in every leave-one-out fit; single seed per cell, so the claim is directional (sign), not the exact exponent.

Table 10: Claim–evidence audit for the current draft.

42/123/456 (retain median ≈ 0.005). These regenerated checkpoints supersede the distributed SalUn baseline; regeneration is driven by `scripts/analysis/37_regen_salun_baselines.sh` (summary in `results/analysis/reports/salun_regen_summary.txt`).

Reconstructed BalDRO baseline; ORBIT/RURK unresolved. The plain BalDRO, ORBIT, and RURK baseline recipes were never committed to version control (only `_smoke` variants exist on any branch), so the originals cannot be regenerated. As a partial check, a BalDRO baseline *reconstructed* by scaling the smoke recipe to the SalUn conventions (objective `bal_dro`, rank 8, 50 epochs, $\mathrm{lr} = 10^{-3}$; `configs/suites/recon_baldro_rurk.yaml`) reproduces proper unlearning across three seeds—clean forget accuracy 0.0%, forget recovery median 0.00417, 0.00355, 0.00447 for seeds 42/123/456. This is a *reconstruction*, not the original baseline, and is not expected to match the original Table 3 numbers; it is reported only to confirm the objective and pipeline behave sensibly. The RURK reconstruction was not completed (its objective's

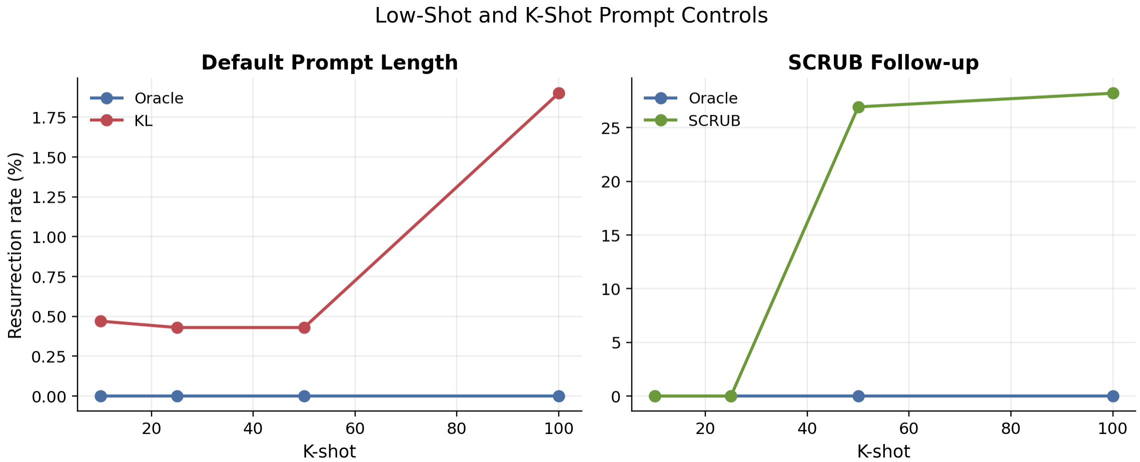

Figure 10: K-shot prompt controls. Oracle stays flat, KL remains weakly vulnerable, and SCRUB only becomes unstable in the larger-shot follow-up.

nested noisy forward made training prohibitively slow), and ORBIT has no recoverable recipe or objective implementation; both remain single-run artifacts not yet independently reproduced (`scripts/analysis/38_recon_baldro_rurk.sh`).

I Appendix: Adaptation Protocol

A critical caveat for the fairness of this benchmark concerns CE-U, SGA, BalDRO, and RURK. These methods were originally proposed in the context of Large Language Models (LLMs) or against adversarial threat models. The implementations evaluated in this paper are *objective-level adaptations* to the vision domain (CIFAR-10 classification with ViT architectures), not faithful reproductions of their original settings or training dynamics. They are included to provide a diverse set of algorithmic loss structures for the robustness microscope to probe, not to formally evaluate the original authors' claims in the domain they were intended for.

Concretely, the adaptations differ from their sources as follows, and these fidelity labels appear in Table 1. **CE-U** is implemented as cross-entropy *ascent* (negative CE on the forget class)—the gradient-ascent objective that [11] was specifically designed to *replace* with a stable corrected-target cross-entropy—so the CE-U row is a gradient-ascent baseline, not a reproduction of CE-U. **RURK**'s robustness term ascends on *random* Gaussian-noise copies of forget inputs, whereas [6] targets robustness to *adversarial*/worst-case perturbed samples; random-noise robustness is strictly weaker. **BalDRO** is a softmax reweighting of hard forget samples, not the min–max distributionally robust optimization with the balancing constraint of [9]. **SGA** (label-smoothed gradient ascent) is the closest to its source [8]. We keep the original method names as row labels for traceability, but read them through these fidelity caveats rather than as faithful baselines.

J Appendix: Low-Shot Controls

The lower-shot prompt controls help separate real recovery from trivial prompt memorization. Figure 10 summarizes two useful trends: oracle controls remain flat near zero across the tracked shot counts, while KL and SCRUB can become vulnerable only in specific low-shot or high-shot regimes.

References

[1] Meghdad Kurmanji, Peter Triantafillou, Jamie Hayes, and Eleni Triantafillou. Towards unbounded machine unlearning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[2] Chongyu Fan, Jiancheng Liu, Yihua Zhang, Eric Wong, Dennis Wei, and Sijia Liu. SalUn: Empowering machine unlearning via gradient-based weight saliency in both image classification and generation. In *International Conference on Learning Representations (ICLR)*, 2024.

[3] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022.

[4] Francesco Croce and Matthias Hein. Reliable evaluation of adversarial robustness with an ensemble of diverse parameter-free attacks. In *International Conference on Machine Learning (ICML)*, 2020.

[5] Menglin Jia, Luming Tang, Bor-Chun Chen, Claire Cardie, Serge Belongie, Bharath Hariharan, and Ser-Nam Lim. Visual prompt tuning. In *European Conference on Computer Vision (ECCV)*, 2022.

[6] Hsiang Hsu, Pradeep Niroula, Zichang He, Ivan Brugere, Freddy Lecue, and Chun-Fu Chen. The unseen threat: Residual knowledge in machine unlearning under perturbed samples. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[7] Weijia Shi, Jaechan Lee, Yangsibo Huang, Sadhika Malladi, Jieyu Zhao, Ari Holtzman, Daogao Liu, Luke Zettlemoyer, Noah A. Smith, and Chiyuan Zhang. MUSE: Machine unlearning six-way evaluation for language models. *arXiv preprint arXiv:2407.06460*, 2024.

[8] Zirui Pang, Hao Zheng, Zhijie Deng, Ling Li, Zixin Zhong, and Jiaheng Wei. Label smoothing improves gradient ascent in LLM unlearning. *arXiv preprint arXiv:2510.22376*, 2025.

[9] Pengyang Shao, Naixin Zhai, Lei Chen, Yonghui Yang, Fengbin Zhu, Xun Yang, and Meng Wang. BalDRO: A distributionally robust optimization based framework for large language model unlearning. *arXiv preprint arXiv:2601.09172*, 2026.

[10] Eli Chien, Haoyu Wang, Ziang Chen, and Pan Li. Langevin unlearning: A new perspective of noisy gradient descent for machine unlearning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[11] Bo Yang. CE-U: Cross entropy unlearning. *arXiv preprint arXiv:2503.01224*, 2025.

[12] Liheng Yu, Zhe Zhao, Yuxuan Wang, Pengkun Wang, Xiaofeng Cao, Binwu Wang, and Yang Wang. FaLW: A forgetting-aware loss reweighting for long-tailed unlearning. In *International Conference on Learning Representations (ICLR)*, 2026.

[13] Jeremy Cohen, Elan Rosenfeld, and J. Zico Kolter. Certified adversarial robustness via randomized smoothing. In *International Conference on Machine Learning (ICML)*, 2019.

[14] Zijie Zhang, Yang Zhou, Xin Zhao, Tianshi Che, and Lingjuan Lyu. Prompt certified machine unlearning with randomized gradient smoothing and quantization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[15] Hao Xuan and Xingyu Li. Verifying robust unlearning: Probing residual knowledge in unlearned models. *arXiv preprint arXiv:2504.14798*, 2025.

[16] Anmol Goel, Alan Ritter, and Iryna Gurevych. Auditing language model unlearning via information decomposition. *arXiv preprint arXiv:2601.15111*, 2026.

[17] SeungBum Ha, Saerom Park, and Sung Whan Yoon. Unlearning's blind spots: Over-unlearning and prototypical relearning attack. *arXiv preprint arXiv:2506.01318*, 2025.

[18] Zhenyu Yu, Yangchen Zeng, Chunlei Meng, Guangzhen Yao, and Shuigeng Zhou. Can vision models truly forget? Mirage: Representation-level certification of visual unlearning. *arXiv preprint arXiv:2605.20282*, 2026.

[19] Carolin Heinzler, Kasra Malihi, and Amartya Sanyal. Less noise, same certificate: Retain sensitivity for unlearning. *arXiv preprint arXiv:2603.03172*, 2026.

[20] Siqiao Mu and Diego Klabjan. Rewind-to-delete: Certified machine unlearning for nonconvex functions. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[21] Anastasia Koloskova, Youssef Allouah, Animesh Jha, Rachid Guerraoui, and Sanmi Koyejo. Certified unlearning for neural networks. *arXiv preprint arXiv:2506.06985*, 2025.

[22] Lulu Xue, Shengshan Hu, Wei Lu, Yan Shen, Dongxu Li, Peijin Guo, Ziqi Zhou, Minghui Li, Yanjun Zhang, and Leo Yu Zhang. Towards reliable forgetting: A survey on machine unlearning verification. *ACM Computing Surveys*, 2026.

[23] Aaradhya Pandey and Sanjeev Kulkarni. Statistical unlearning of distributions: A hypothesis testing approach. *arXiv preprint arXiv:2605.16645*, 2026.

[24] Jiatong Yu, Yinghui He, Anirudh Goyal, and Sanjeev Arora. On the impossibility of retrain equivalence in machine unlearning. *arXiv preprint arXiv:2510.16629*, 2025.

[25] Ali Ebrahimpour-Boroojeny, Hari Sundaram, and Varun Chandrasekaran. AMUN: Adversarial machine UNlearning. *arXiv preprint arXiv:2503.00917*, 2025.